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Ultra-Fusion: A Resilient Tightly-Coupled Multi-Sensor Fusion SLAM Framework under Sensor Degradation and Spatiotemporal Perturbation for Intelligent Transportation Systems

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Project webpage: https://sjtuyinjie.github.io/ultrafusion-web/

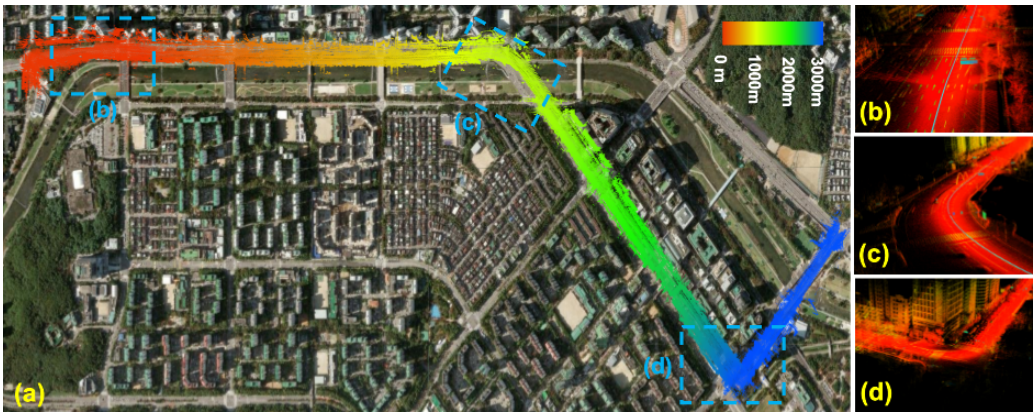

Fig. 1. Ultra-Fusion supports road-level localization and reconstruction for urban autonomous driving. (a) A 3 km KAIST urban-driving route, color-coded by traveled distance. (b)–(d) Local reconstructions along the route, preserving road boundaries, roadside structures, and the estimated vehicle trajectory (blue).

***Abstract*—Reliable localization is essential for intelligent transportation systems (ITS), including autonomous vehicles, quadruped last-mile carriers, and infrastructure-inspection unmanned aerial vehicles (UAVs). Although tightly-coupled multi-sensor fusion improves accuracy in favorable conditions, deployed systems remain vulnerable to sensor degradation—poor illumination, LiDAR degeneracy, wheel slippage, and GNSS outage—and to spatiotemporal calibration errors. These failures are common in urban canyons, tunnels, and high-speed corridors, where localization drift can degrade route tracking, tunnel passage continuity, and local map alignment. This paper presents Ultra-Fusion, a tightly-coupled multi-sensor localization framework based on a unified sliding-window estimator. Asynchronous measurements are timestamp-ordered and converted into optional factors within one optimization window, supporting WIO, VIO, LIO, and LVIO with optional wheel and GNSS augmentation. Observability-aware initialization selects the bootstrap mode, factor-wise reliability scheduling gates degraded measurements, and online LiDAR–IMU spatiotemporal calibration refines temporal offsets and rotational extrinsics during operation. We extend the M3DGR benchmark with simulation trajectories and evaluate more than 60 open-source SLAM systems on M3DGR, M2DGR-Plus, KAIST, GrandTour, and MARS-LVIG. The results show competitive accuracy across wheeled, legged, and aerial platforms under long-duration and high-speed operation, degradation, and calibration perturbation, improving localization availability for road-level autonomy, campus and warehouse mobility, and low-altitude aerial inspection. To benefit the industrial and academic community, we will release source code and datasets upon paper acceptance.**

***Index Terms*—System State Estimation, Sensor Technology, Connected and Autonomous Vehicles, Autonomous Driving, Multi-sensor Fusion, System Design**

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I. Introduction

RELIABLE localization is central to intelligent transportation systems, including autonomous vehicles, advanced driver-assistance systems (ADAS), infrastructure-assisted mobility, quadruped delivery and inspection platforms, and low-altitude UAV operations [1], [2]. In urban streets, highways, tunnels, campuses, and air corridors, localization must handle changing sensing conditions, platform motions, and environmental uncertainty. SLAM has therefore evolved from single-modality pipelines toward multi-sensor fusion with LiDAR, cameras, GNSS, wheel odometry, and IMUs. Although existing tightly- and loosely-coupled frameworks perform well in structured settings [3]–[5], robustness across sensor configurations, platforms, and operating conditions remains challenging.

This gap reflects several coupled requirements. First, ITS fleets rarely share the same sensor suite: road vehicles and shuttles may rely on wheel odometry and intermittent GNSS, quadruped platforms encounter slip and body oscillation on uneven terrain, and UAVs face weak structure at altitude, whereas each may combine different visual, LiDAR, and inertial sensors. Fixed-stack systems often require changes to state definitions, initialization, factor activation, calibration variables, or marginalization when the sensor suite changes. Second, sensor reliability varies over time. Poor illumination, dynamic occlusion, motion blur, adverse weather, LiDAR degeneracy, wheel slippage, and GNSS denial can make useful measurements unreliable; fixed-confidence fusion may then bias the optimization or cause tracking loss. Third, spatiotem-

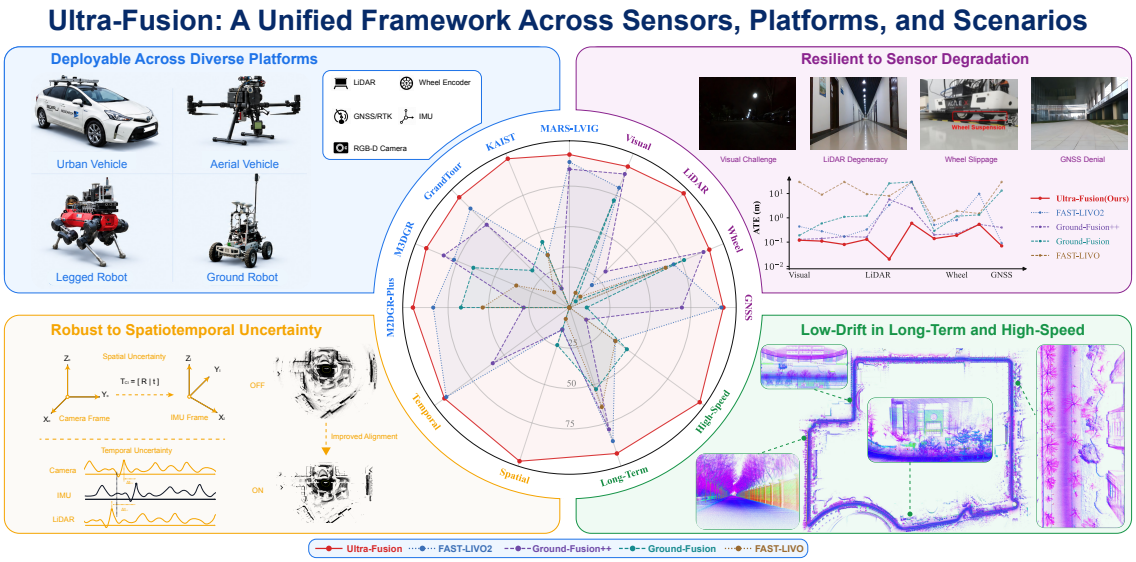

Fig. 2. Overview of Ultra-Fusion as a unified multi-sensor SLAM framework across sensors, platforms, and scenarios. The framework supports heterogeneous inputs, deploys on ground, aerial, legged, and vehicle platforms, and improves localization robustness under sensor degradation, spatiotemporal uncertainty, and long-term or high-speed operation.

poral inconsistency is common in multi-sensor platforms, especially in LVIO, where time offsets and extrinsic errors distort cross-modal correspondences and accumulate drift [6], [7]. Based on these observations, we argue that ITS localization should be treated as a unified estimation problem rather than as configuration-specific pipelines. A desirable framework should support multiple sensor configurations, initialize under sufficient observability, adapt degraded factors online, and refine calibration only under reliable excitation. Evaluation should also go beyond a single complete sensor suite, since robustness depends on whether each configuration remains competitive within its method category and transfers across representative platforms and scenarios.

To address these requirements, we propose Ultra-Fusion, a tightly-coupled multi-sensor framework for ITS localization under flexible sensor availability. A *Unified Sliding-Window Estimator* orders heterogeneous observations by timestamp and converts them into compatible factors in one optimization window, supporting WIO, VIO, LIO, and LVIO with optional wheel and/or GNSS augmentation. *Observability-Aware Initialization* selects the start-up regime, *Factor-Wise Reliability Scheduling* gates or down-weights unreliable residuals, and *Online Spatiotemporal Calibration* refines temporal and extrinsic parameters under sufficient excitation and sensor reliability.

Ultra-Fusion provides a **unified framework across sensors, platforms, and scenarios**: WIO, VIO, LIO, LVIO and their variants share initialization, reliability scheduling, calibration, and marginalization logic. We evaluate the estimator under different sensor availability, platform dynamics, and operating conditions, including simulation-based perturbations and an expanded M3DGR comparison. The main contributions are:

- We propose **Ultra-Fusion**[1], a tightly-coupled multi-sensor fusion framework for intelligent transportation systems. A unified sliding-window estimator with observability-aware initialization supports WIO, VIO, LIO, and LVIO within one configurable optimization framework that shares state representation, factor admission, calibration, and marginalization.
- We develop **Factor-Wise Reliability Scheduling** to address sensor degradation during online estimation. The scheduler applies degeneracy-aware gating and down-weighting directly within the unified optimization problem, improving robustness under poor illumination, LiDAR degeneracy, wheel slippage, and GNSS denial.
- We design **Online Spatiotemporal Calibration** to handle calibration uncertainty during operation. The calibration variables are refined only under sufficient excitation and sensor reliability, reducing cross-modal bias caused by temporal offsets and extrinsic perturbations.
- We extend the **M3DGR benchmark**[2] and conduct a comprehensive large-scale evaluation of **more than 60** representative SLAM systems, including controlled simulation-based perturbation studies. This benchmark study provides a systematic analysis of robustness trends, degradation patterns, and common failure modes.
- We validate Ultra-Fusion on heterogeneous ITS-relevant platforms and operating conditions, including autonomous driving, campus and warehouse wheeled robots, quadruped robots, and inspection UAVs, as well as long-term and high-speed trajectories.

Together, the framework and benchmark support deployment-oriented multi-sensor localization under sensing degradation and calibration uncertainty across multimodal transportation platforms.

[1] https://github.com/sjtuyinjie/Ultra-Fusion

[2] https://github.com/sjtuyinjie/M3DGR

TABLE I. COMPARISON OF REPRESENTATIVE MULTI-SENSOR FUSION SLAM SYSTEMS, EMPHASIZING CONFIGURABILITY AND ROBUSTNESS.

Method/Year	Sensor Set[1]	Configurable Modes	Tightly-Coupled	Degradation-aware	Online Calibration	Mapping[2]
VINS-RGBD [8], 2019	CID	Fixed	✓			SPC/Non-color
DRE-SLAM [9], 2019	CDW	Fixed	✓			Mesh/Non-color
GR-Fusion [10], 2021	CIWGL	Fixed	✓			SPC/Non-color
LVI-SAM [11], 2021	CIL	Fixed				SPC/Non-color
VIW-Fusion [12], 2022	CIW	Fixed	✓		Spatial+Temporal	SPC/Non-color
R3LIVE [3], 2022	CIL	Fixed	✓			**Mesh/Color**
DAMS-LIO [13], 2023	IWL	Fixed	✓	L		SPC/Non-color
M2C-GVIO [14], 2023	CIG	Fixed	✓			Sparse/Non-color
FAST-LIVO2 [15], 2024	CIL	Fixed	✓	L		*DPC/Color*
Ground-Fusion [7], 2024	CIDWG	GNSS-optional	✓	CDWG	Spatial	*DPC/Color*
LIGO [16], 2025	IGL	Fixed	✓			SPC/Non-color
Super Odometry [17], 2025	CIL	Fixed	✓	L		SPC/Non-color
Ground-Fusion++ [18], 2025	CIDWGL	GNSS-optional		L		**Mesh/Color**
Ultra-Fusion (Ours), 2026	CIDWGL	**WIO & VIO/LIO/LVIO + optional W/G**	✓	CIDWGL	Spatial+Temporal	**GS/Color**

[1] C: RGB camera, I: IMU, D: depth camera, W: wheel odometry, G: GNSS, L: LiDAR.[2] SPC: Sparse Point Cloud, DPC: Dense Point Cloud, GS: Gaussian Splatting.

II. RELATED WORK

A. Multi-sensor Fusion SLAM

Multi-sensor fusion SLAM exploits complementary measurements to balance accuracy, drift, and robustness. Visual and visual–inertial systems are effective in texture-rich scenes [19]–[21], while LiDAR-centric pipelines provide geometric observability and scale consistency [22]–[24]. Recent LiDAR–visual–inertial systems combine these cues for pose estimation and mapping [3], [15], [25]. In ITS, road vehicles, quadruped carriers, and inspection UAVs often add wheel odometry and GNSS for short-term motion constraints and long-term drift correction [12], [14], [16], [26], [27]. A key design choice is how LiDAR geometry enters the estimator. Most LiDAR-aided systems use scan-to-map registration or a LiDAR odometry thread [15], [22]–[24], [28], [29]. Some continuous-time systems expose raw or point-level LiDAR residuals to a bounded optimization window [30]–[32]. These methods motivate direct LiDAR factors for asynchronous acquisition and motion distortion, but are usually limited to LIO/LIC settings or dedicated continuous-time trajectory parameterizations.

Many multi-sensor systems remain tied to fixed sensor stacks or coupled subsystems [7], [9], [11]. Changing the deployment configuration may affect the state definition, factor activation, calibration handling, and marginalization interface. Ultra-Fusion addresses this gap with a *Unified Sliding-Window Estimator*, where LiDAR residuals, visual reprojection factors, inertial constraints, wheel factors, and GNSS anchors share one state, reliability scheduler, and initialization strategy. This retains direct LiDAR factors while supporting WIO, VIO, LIO, LVIO and their variants in one estimator. Table I summarizes representative fusion systems.

B. Sensor Degradation and Spatiotemporal Miscalibration

Sensor degradation is a common source of SLAM performance loss. Illumination variation, motion blur, dynamic occlusion, geometric degeneracy, wheel slip, or satellite blockage reduce reliable observations and increase outliers; without reliability control, they may cause tracking loss or drift. Degeneracy-aware SLAM methods adapt the estimator through gating, down-weighting, or modality switching [4], [7], [13]. Ground-Fusion [7] and Ground-Fusion++ [18] demonstrate reliability-conditioned fusion [33], but remain tied to limited modalities or subsystem-level decisions. Ultra-Fusion applies reliability control directly to LiDAR, vision, IMU, wheel, and GNSS factors inside the common optimization problem.

Spatiotemporal miscalibration is another source of bias. Time offsets may arise from asynchronous triggering, latency, or imperfect timestamping, while extrinsics may change with vibration, temperature, or reconfiguration; in large-scale ITS platforms, small errors can bias cross-modal alignment and accumulate drift. Prior work studies joint spatial–temporal calibration [34], [35] and calibration toolboxes [36], but these methods are usually offline or pre-deployment. Ultra-Fusion refines time-offset and extrinsic variables inside the SLAM loop when observability and sensor reliability are sufficient.

C. SLAM Benchmark Datasets

Benchmark datasets are essential for reproducible SLAM evaluation. For ITS-oriented research, an informative benchmark should cover transportation scenarios, diverse sensor configurations, degradation cases, and trustworthy ground truth. Existing datasets often cover only part of this scope: some focus on limited modalities, some provide mild environmental variation, and others rely on proxy trajectories that may be unreliable in difficult sequences. For example, EuRoC and TUM-VI mainly support visual–inertial evaluation in relatively controlled environments [37], [38], while datasets such as OpenLORIS-Scene and Ground-Challenge broaden the sensing setup but still have limited coverage of systematic degradation conditions [39], [40]. More recent datasets expand modality and scenario diversity for road, aerial, and non-road mobility [7], [41]–[44], yet large-scale stress testing under controlled degradation and calibration perturbation remains relatively underexplored.

This motivates benchmarks with rich sensing streams and systematic robustness evaluation [4], [7], [14], [16], [45]. We present M3DGR, a sensor-rich benchmark for staged degradation, including visibility challenges, geometric degeneracy, wheel slip, and GNSS denial [18]. M3DGR provides synchronized and calibrated sensor streams for analyzing degradation robustness and spatiotemporal consistency. On this benchmark, we evaluate **over 60** representative SLAM systems and analyze common failure modes.

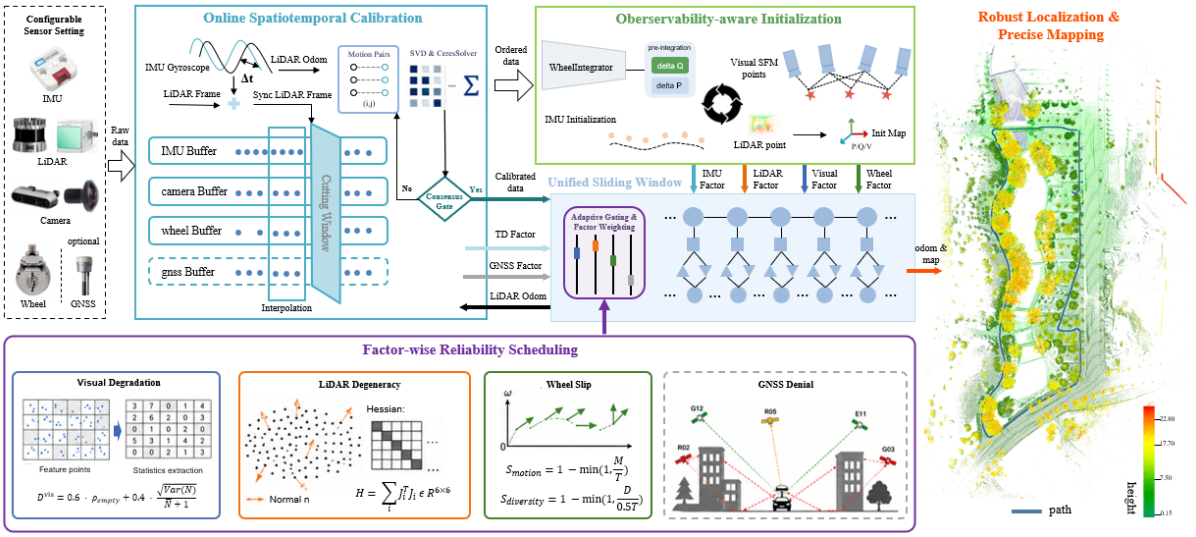

Fig. 3. Overview of the Ultra-Fusion framework. Configurable sensor streams are timestamp-ordered, calibrated online, and initialized under observability-aware bootstrap before entering a unified sliding-window factor graph. Factor-wise reliability scheduling suppresses degraded measurements, while retained factors are jointly optimized across WIO, VIO, LIO, and LVIO configurations with shared state, marginalization, optional wheel/GNSS augmentation, and online calibration.

III. Methodology

Building on our IROS 2025 Ground-Fusion++ system [18], Ultra-Fusion reorganizes the conference baseline into a journal-ready ITS localization framework through five extensions: (i) a unified optional-factor sliding-window estimator rather than subsystem coupling, (ii) observability-aware initialization, (iii) in-graph factor-wise reliability scheduling, (iv) online LiDAR–IMU spatiotemporal calibration, and (v) an expanded M3DGR benchmark with simulation perturbations and cross-platform validation on KAIST, GrandTour, and MARS-LVIG.

Unlike scan-to-map LIO/LVIO pipelines that inject LiDAR odometry as an external prior [15], [22]–[24], and unlike Ground-Fusion++ where reliability and calibration remain largely subsystem-specific, Ultra-Fusion keeps LiDAR geometric residuals together with visual, inertial, wheel, and GNSS factors in one shared sliding window. Compared with full continuous-time LIO/LIC systems [30]–[32], [46], this retains point-level LiDAR constraints while preserving a compact optimization structure. As shown in Fig. 3, timestamp-ordered measurements pass through online calibration and observability-aware bootstrap before entering the factor graph, where reliability scheduling suppresses unreliable modalities and the active window jointly optimizes retained factors. The following subsections present state representation, the unified estimator, initialization, reliability scheduling, online calibration, and mapping-oriented extensions.

A. State Representation and Temporal Ordering

At timestamp k, the platform state is

$$\boldsymbol{x}_k \triangleq \{\mathbf{R}_k,\ \boldsymbol{t}_k,\ \boldsymbol{v}_k,\ \boldsymbol{b}_{a,k},\ \boldsymbol{b}_{g,k}\}, \tag{1}$$

where $\mathbf{R}_k \in SO(3)$, $\boldsymbol{t}_k, \boldsymbol{v}_k \in \mathbb{R}^3$, and $\boldsymbol{b}_{a,k}, \boldsymbol{b}_{g,k} \in \mathbb{R}^3$. The active window $\mathscr{X} = \{\boldsymbol{x}_{k-W+1}, \ldots, \boldsymbol{x}_k\}$ receives timestamp-ordered constraints from all available sensors. LiDAR extrinsics are $\mathbf{T}_{IL} = \{\mathbf{R}_{IL}, \boldsymbol{t}_{IL}\}$, while camera/wheel extrinsics and temporal offsets are

$$\mathbf{T}_{IC} = \{\mathbf{R}_{IC}, \boldsymbol{t}_{IC}\}, \quad \mathbf{T}_{IO} = \{\mathbf{R}_{IO}, \boldsymbol{t}_{IO}\}, \quad t_d^C, \quad t_d^O, \quad \Delta t_{LI}, \tag{2}$$

where t_d^C, t_d^O, and Δt_{LI} denote camera–IMU, wheel–IMU, and LiDAR–IMU time offsets, respectively.

Although the estimator is keyframe-indexed, intra-frame measurements query a local continuous-time segment. For a LiDAR point acquired at normalized time $\alpha_i \in [0,1]$, the deskewing pose is

$$\mathbf{R}_k(\alpha_i) = \mathrm{Slerp}(\mathbf{R}_k^b, \mathbf{R}_k^e; \alpha_i), \quad \boldsymbol{t}_k(\alpha_i) = (1-\alpha_i)\boldsymbol{t}_k^b + \alpha_i \boldsymbol{t}_k^e, \tag{3}$$

where superscripts b and e denote scan begin and end poses. This preserves point-time coupling [30], [31], [46] without introducing a separate batch trajectory. Pose variables are updated on the manifold using a right-multiplicative local parameterization:

$$\begin{aligned} \mathbf{R}_k &\leftarrow \mathbf{R}_k \mathrm{Exp}([\delta\boldsymbol{\theta}_k]_\times), \\ \boldsymbol{t}_k &\leftarrow \boldsymbol{t}_k + \delta\boldsymbol{t}_k, \end{aligned} \tag{4}$$

with $[\cdot]_\times$ the skew operator and $\mathrm{Exp}: \mathbb{R}^3 \to SO(3)$.

B. Unified Sliding-Window Estimator

Ultra-Fusion represents heterogeneous measurements as optional factors sharing one state, prior, calibration parameterization, and reliability scheduler. The core LiDAR–IMU objective is

$$\begin{aligned} \min_{\mathscr{X}} \quad & \mathscr{L}_{\text{prior}} \\ & + \sum_{k' \in \mathscr{W}} \left(\mathscr{L}_{\text{lidar}}(k') + \lambda_{\text{imu}} \mathscr{L}_{\text{imu}}(k') \right), \end{aligned} \tag{5}$$

where $\mathscr{W}$ is the active window index set, $\lambda_{\text{imu}} \in [0,1]$ balances inertial coupling, $\rho(\cdot)$ denotes a robust loss, and $\boldsymbol{\Omega}_{(\cdot)}$ denotes the information matrix of the corresponding factor.

Available visual, LiDAR, wheel, and GNSS measurements instantiate additional factors, enabling WIO, VIO, LIO, LVIO, and augmented variants without changing the state or marginalization interface. Unlike subsystem-level coordination [18], robustness is controlled at factor level through activation, suppression, or down-weighting.

LiDAR geometric factor. LiDAR supplies geometric constraints in large-scale transportation scenes. Unlike scan-to-map pipelines that use point-to-plane registration to produce a LiDAR odometry update outside the multi-sensor window [22]–[24], [28], Ultra-Fusion keeps the point-to-plane residual inside the shared objective. For a surface point $\boldsymbol{p}_i^L$ acquired at intra-scan time α_i, the transformed point $\boldsymbol{p}_i^w$ is constrained by its matched local plane $\{\boldsymbol{n}_i, \boldsymbol{s}_i\}$:

$$\boldsymbol{p}_i^w = \mathbf{R}_k(\alpha_i)(\mathbf{R}_{IL}\boldsymbol{p}_i^L + \boldsymbol{t}_{IL}) + \boldsymbol{t}_k(\alpha_i), \quad r_i^{\text{lidar}} = \boldsymbol{n}_i^\top(\boldsymbol{p}_i^w - \boldsymbol{s}_i). \tag{6}$$

The corresponding robust factor is

$$\mathscr{L}_{\text{lidar}} = \sum_{i\in\mathscr{S}_k} \rho\left(\omega_i (r_i^{\text{lidar}})^2\right). \tag{7}$$

Here $\mathscr{S}_k$ is the accepted surface-feature set and ω_i encodes local plane uncertainty. Map-neighborhood queries reuse efficient scan-to-map geometry [23], [24], [29], while optional intensity consistency provides auxiliary constraints in geometrically weak regions when sufficient support exists.

IMU preintegration factor. To preserve short-term motion consistency between geometric updates, Ultra-Fusion keeps a bias-corrected inertial bridge between adjacent window states [21], [47]. We denote the compact preintegration residual by $\boldsymbol{r}_{\text{imu}}$ and write

$$\mathscr{L}_{\text{imu}} := \rho\left(\left\|\boldsymbol{\Omega}_{\text{imu}}^{1/2}\boldsymbol{r}_{\text{imu}}(\boldsymbol{x}_{k-1}, \boldsymbol{x}_k, \boldsymbol{b}_a, \boldsymbol{b}_g)\right\|^2\right). \tag{8}$$

The contribution of this factor is controlled by the IMU reliability score and the coupling weight λ_{imu}.

Wheel preintegration factor. Wheel odometry provides planar motion constraints but is sensitive to slip and kinematic mismatch. We model it as a preintegrated relative-motion factor with IMU-to-wheel extrinsic $\mathbf{T}_{IO}$, wheel scale $\boldsymbol{s}$, and temporal compensation [12], [48]:

$$\mathscr{L}_{\text{wheel}} := \rho\left(\left\|\boldsymbol{\Omega}_{\text{wheel}}^{1/2}\boldsymbol{r}_{\text{wheel}}(\boldsymbol{x}_{k-1}, \boldsymbol{x}_k, \mathbf{T}_{IO}, \boldsymbol{s}, t_d^O)\right\|^2\right). \tag{9}$$

Its influence is governed by wheel-slip consistency, avoiding over-constraint of weakly observable directions.

Visual reprojection factor. Visual measurements enter through a temporally compensated reprojection residual using $\mathbf{T}_{IC}$, inverse depth λ_{k-1}, and offset t_d^C [3], [15], [21]:

$$\boldsymbol{r}^{\text{vis}} := \pi\left(\mathbf{T}_{IC}^{-1}\mathbf{T}_k^{-1}\mathbf{T}_{k-1}\mathbf{T}_{IC}\left(\frac{1}{\lambda_{k-1}}\tilde{\boldsymbol{u}}_{k-1,t_d^C}\right)\right) - \tilde{\boldsymbol{u}}_{k,t_d^C}, \tag{10}$$

$$\mathscr{L}_{\text{vis}} = \rho\left(\|\boldsymbol{\Omega}_{\text{vis}}^{1/2}\boldsymbol{r}^{\text{vis}}\|^2\right). \tag{11}$$

Here $\mathbf{T}_{k-1}$ and $\mathbf{T}_k$ are platform poses, $\tilde{\boldsymbol{u}}_{\ell,t_d^C}$ is a time-compensated normalized image point, and $\pi(\cdot)$ is the normalized-plane projection. Track age, spatial distribution, KLT consistency, and epipolar checks provide reliability evidence for scheduling.

GNSS position anchoring factor. When available, GNSS provides integrity-checked global constraints for drift suppression [14], [16], [26]. For compactness, we first write the optional position anchoring factor:

$$\mathscr{L}_{\text{gnss}} := \rho\left(\left\|\boldsymbol{\Omega}_{\text{gnss}}^{1/2}(\boldsymbol{t}_k - \boldsymbol{p}_k^{\text{gnss}})\right\|^2\right), \tag{12}$$

where $\boldsymbol{p}_k^{\text{gnss}}$ is the GNSS position measurement expressed in the estimator frame, and the measurement covariance and factor activation are governed by the GNSS integrity checks in the reliability scheduler. Raw pseudorange/Doppler measurements are handled by separate factors with receiver clock bias/drift, an ECEF anchor, and ENU–local yaw alignment, and are admitted by the same integrity gates.

After assembling active geometric, inertial, calibration, and reliability-gated factors, Ultra-Fusion solves a robust nonlinear least-squares problem. Historical information is retained by a Gaussian marginalization prior,

$$\mathscr{L}_{\text{prior}} := \left\|\boldsymbol{\Omega}_{\text{prior}}^{1/2}\boldsymbol{r}_{\text{prior}}(\mathscr{X})\right\|^2, \tag{13}$$

where $\boldsymbol{r}_{\text{prior}}$ is obtained by Schur-complement or QR marginalization [20], [21]. The prior remains compatible across sensor configurations, while eigenvalue truncation and conservative fallback settings improve numerical stability when active factors change.

C. Observability-Aware Initialization

A well-constrained initial state is required before activating the unified sliding-window estimator. We formulate initialization as observability-aware model selection: motion excitation and sensing geometry determine whether the estimator uses SfM-based visual–inertial alignment, stationary or wheel-aided inertial alignment, or LiDAR-odometry-aided short-window MAP estimation; otherwise, the initialization window is extended until sufficient evidence is available.

The resulting bootstrap mode is represented by

$$\rho_{\text{boot}} \in \{\mathsf{D}, \mathsf{S}, \mathsf{M}, \mathsf{A}\}, \tag{14}$$

where D denotes the dynamic visual–inertial hypothesis, S denotes the stationary or wheel-aided inertial hypothesis, M denotes the LiDAR-odometry-aided MAP hypothesis, and A denotes deferred initialization under insufficient observability.

In Algorithm 1, E_ω and E_v quantify motion excitation, $\bar{N}^{\text{feat}}$ and $\bar{p}$ quantify visual support, and N^{lidar} measures valid LiDAR geometric support. The bootstrap mode ρ_{boot} denotes the selected hypothesis, while $\pi_{\text{boot}} = 1$ admits initialization after the MCC consistency check. Under the LiDAR-odometry-aided hypothesis (M), gravity-aligned scan-matching poses provide geometric priors for a short-window MAP estimate of velocity and IMU biases.

D. Factor-Wise Reliability Scheduling

Reliability scheduling prevents degraded residuals from dominating the optimizer. At each keyframe, modality-specific degeneracy scores $D_k^{(m)}$ are computed from the evidence below, mapped to binary activation variables $s_k^{(m)}$ and, when appropriate, covariance inflation, smoothed with short-horizon hysteresis to avoid frequent switching, and incorporated into the unified sliding-window objective without altering the state definition or marginalization interface.

At time k, each modality $m \in \{\text{LiDAR}, \text{Visual}, \text{IMU}, \text{Wheel}, \text{GNSS}\}$ is assigned a normalized degeneracy score $D_k^{(m)} \in [0,1]$ and an activation indicator

$$s_k^{(m)} = \mathbf{1}\left[D_k^{(m)} \le \tau^{(m)} \ \wedge \ N_k^{(m)} \ge N_{\min}^{(m)}\right], \tag{15}$$

where $N_k^{(m)}$ is the valid observation count. The scheduled objective in the sliding window is

$$\begin{aligned}\min_{\mathscr{X}} \quad & \mathscr{L}_{\text{prior}} + \sum_{k'\in\mathscr{W}} \Big(s_{k'}^{(\text{LiDAR})}\mathscr{L}_{\text{lidar}}(k') + \lambda_{\text{imu}}\, s_{k'}^{(\text{IMU})}\mathscr{L}_{\text{imu}}(k') \\ & + s_{k'}^{(\text{Visual})}\mathscr{L}_{\text{vis}}(k') + s_{k'}^{(\text{Wheel})}\mathscr{L}_{\text{wheel}}(k') + s_{k'}^{(\text{GNSS})}\mathscr{L}_{\text{gnss}}(k')\Big).\end{aligned} \tag{16}$$

Algorithm 1 Observability-Aware Initialization

Require: Initial buffer $\mathscr{W}_{\text{init}}$, IMU segment $\mathscr{I}_0$, thresholds $\{\tau_\omega, \tau_v, \tau_p, \tau_\omega^\Sigma, \tau_a^\Sigma, N_{\min}^{\text{feat}}, N_{\min}^{\text{lidar}}\}$
Ensure: Initial ESKF state $\boldsymbol{x}_0^{\text{eskf}}$, bootstrap mode ρ_{boot}, and admission indicator $\pi_{\text{boot}} \in \{0,1\}$

Compute IMU statistics $(\bar{\boldsymbol{a}}, \bar{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_a, \boldsymbol{\Sigma}_\omega)$ and gravity-aligned attitude seed $\mathbf{R}_0$
Evaluate excitation, visual, wheel, and LiDAR-geometry indicators $(E_\omega, E_v, \bar{N}^{\text{feat}}, \bar{p}, N^{\text{lidar}})$ on $\mathscr{W}_{\text{init}}$
if $E_\omega > \tau_\omega$ **and** $\bar{N}^{\text{feat}} \geq N_{\min}^{\text{feat}}$ **and** $\bar{p} \geq \tau_p$ **then**
  $\rho_{\text{boot}} \leftarrow \mathsf{D}$ (SfM-based visual–inertial branch)
  Run SfM + visual–inertial alignment to recover $\{s, \boldsymbol{g}, \boldsymbol{v}_{0:W}, \boldsymbol{b}_g\}$
  Repropagate IMU preintegration with updated bias
else if $E_\omega \leq \tau_\omega$ **and** $E_v \leq \tau_v$ **and** $\|\boldsymbol{\Sigma}_\omega\|_F \leq \tau_\omega^\Sigma$ **and** $\|\boldsymbol{\Sigma}_a\|_F \leq \tau_a^\Sigma$ **then**
  $\rho_{\text{boot}} \leftarrow \mathsf{S}$ (stationary/wheel-aided inertial branch)
  Set inertial seed: $\boldsymbol{b}_g^0 \leftarrow \bar{\boldsymbol{\omega}}$, $\boldsymbol{g}^0 \leftarrow -g_n \bar{\boldsymbol{a}}/\|\bar{\boldsymbol{a}}\|$, $\boldsymbol{b}_a^0 \leftarrow \bar{\boldsymbol{a}} + \mathbf{R}_0^\top \boldsymbol{g}^0$
else if $N^{\text{lidar}} \geq N_{\min}^{\text{lidar}}$ **and** LiDAR geometry check passes **then**
  $\rho_{\text{boot}} \leftarrow \mathsf{M}$ (LiDAR-odometry-aided MAP branch)
  Use scan-matching poses as geometric priors and solve a short-window MAP problem for $\{\boldsymbol{v}_{0:W}, \boldsymbol{b}_a^0, \boldsymbol{b}_g^0\}$
  Repropagate IMU/wheel preintegration with updated bias
else
  $\rho_{\text{boot}} \leftarrow \mathsf{A}$ (deferred initialization)
  Continue data accumulation and return with $\pi_{\text{boot}} = 0$
end if
if $\rho_{\text{boot}} \in \{\mathsf{D}, \mathsf{S}, \mathsf{M}\}$ **and** MCC consistency check passes **then**
  Initialize $\boldsymbol{x}_0^{\text{eskf}} = \{\mathbf{R}_0, \boldsymbol{t}_0, \boldsymbol{v}_0, \boldsymbol{b}_a^0, \boldsymbol{b}_g^0, \boldsymbol{g}^0\}$ and set $\pi_{\text{boot}} = 1$
  Activate LiDAR geometric factors in the unified window
else
  Set $\pi_{\text{boot}} = 0$ and continue accumulation
end if

All scores use the convention that larger values indicate lower reliability, with modality-wise weights normalized. For non-LiDAR modalities, compact consistency scores are

$$\begin{aligned}
D_k^{(\text{Visual})} &= w_f\big(1 - \min(1, N_f/N_f^{\text{ref}})\big) + w_g(1 - G_f) + w_r \min(1, \bar{e}_{\text{repr}}/\tau_r),\\
D_k^{(\text{IMU})} &= w_e(1 - \eta_k^{\text{exc}}) + w_p \min(1, \|\boldsymbol{r}_{\text{imu}}\|^2_{\boldsymbol{\Omega}_{\text{imu}}}/\tau_{\text{imu}}) + w_s \mathbf{1}[\text{saturation}],\\
D_k^{(\text{Wheel})} &= w_v \min(1, \|\Delta\boldsymbol{p}^w - \Delta\boldsymbol{p}^I\|/\tau_v) + w_\psi \min(1, |\Delta\psi^w - \Delta\psi^I|/\tau_\psi),\\
D_k^{(\text{GNSS})} &= w_q(1 - q_{\text{fix}}) + w_\Sigma \min(1, \text{tr}(\boldsymbol{\Sigma}_g)/\tau_\Sigma) + w_v \min(1, \|\boldsymbol{\nu}_g\|^2_{\boldsymbol{\Sigma}_g^{-1}}/\tau_v),
\end{aligned} \tag{17}$$

where N_f, G_f, and $\bar{e}_{\text{repr}}$ denote visual support, $\Delta(\cdot)^w$ and $\Delta(\cdot)^I$ denote wheel- and IMU-predicted increments, and q_{fix}, $\boldsymbol{\Sigma}_g$, and $\boldsymbol{\nu}_g$ denote GNSS integrity evidence. These cues determine factor admission or covariance inflation within the same scheduler.

LiDAR degeneracy detection. LiDAR reliability is evaluated from the point-to-plane Hessian accumulated over the local scan [7], [13]. With Jacobian rows $\boldsymbol{J}_i = [\boldsymbol{n}_i^\top, -\boldsymbol{n}_i^\top [\boldsymbol{p}_i]_\times]$,

$$\mathbf{H}_k = \sum_i \boldsymbol{J}_i^\top \boldsymbol{J}_i + 10^{-8}\mathbf{I}_6, \tag{18}$$

Eigen-decomposition yields $\lambda_1 \leq \cdots \leq \lambda_6$ and $\kappa(\mathbf{H}_k) = \lambda_6/(\lambda_1 + 10^{-12})$. Together with normal covariance $\mathbf{C}_n$ and match count M_k, they define

$$\begin{aligned}
D_k^{(\text{LiDAR})} &= w_h \phi_h(\lambda_1, \kappa) + w_n \frac{\tau_n}{\tau_n + \lambda_3(\mathbf{C}_n)} + w_a \phi_a(\mathbf{H}_k) + w_c\Big(1 - \min\big(1, \tfrac{M_k}{M_{\text{ref}}}\big)\Big),\\
\phi_h(\lambda_1, \kappa) &= \tfrac{1}{2}\Big[\min\big(1, \tfrac{\tau_\lambda}{\lambda_1}\big) + \min\big(1, \tfrac{\kappa}{\tau_\kappa}\big)\Big],
\end{aligned} \tag{19}$$

where $\phi_a(\cdot)$ penalizes axes with weak constraint projection, $\lambda_3(\mathbf{C}_n)$ is the smallest normal-covariance eigenvalue, $\tau_n, \tau_\lambda, \tau_\kappa$ are thresholds, M_{ref} is a reference match count, and w_h, w_n, w_a, w_c are normalized non-negative weights. When this score exceeds the modality-specific threshold, the corresponding LiDAR factors are either deactivated through $s_k^{(\text{LiDAR})}$ or attenuated, thereby preventing ill-conditioned scan matches from degrading the joint solution while retaining informative measurements in well-structured regions.

Visual reliability check. Visual tracking quality varies sharply with illumination and texture. The scheduler combines feature count N_f, spatial distribution uniformity G_f (measured by grid occupancy variance on an 8×8 image partition), forward-backward KLT inlier ratio, and mean reprojection residual $\bar{e}_{\text{repr}}$:

$$D_k^{(\text{Visual})} = w_f\Big(1 - \min\big(1, \tfrac{N_f}{N_f^{\text{ref}}}\big)\Big) + w_g(1 - G_f) + w_r \min\big(1, \tfrac{\bar{e}_{\text{repr}}}{\tau_r}\big), \tag{20}$$

where N_f^{ref} is a reference track count, τ_r a residual threshold, and w_f, w_g, w_r are non-negative weights. Unlike systems that mainly reject measurements before optimization [3], [15], [21], Ultra-Fusion incorporates this evidence into factor scheduling. When $D_k^{(\text{Visual})}$ exceeds its threshold or $N_f < N_{\min}^{(\text{Visual})}$, visual factors are deactivated or attenuated.

IMU excitation consistency. Inertial preintegration supplies short-term motion continuity but can introduce bias under insufficient excitation [21], [47]. The scheduler monitors rotational and translational excitation η_k^{exc}, preintegration residual magnitude $\|\boldsymbol{r}_{\text{imu}}\|^2_{\boldsymbol{\Omega}_{\text{imu}}}$, and saturation flags:

$$D_k^{(\text{IMU})} = w_e(1 - \eta_k^{\text{exc}}) + w_p \min\Big(1, \tfrac{\|\boldsymbol{r}_{\text{imu}}\|^2_{\boldsymbol{\Omega}_{\text{imu}}}}{\tau_{\text{imu}}}\Big) + w_s \mathbf{1}[\text{saturation}], \tag{21}$$

where τ_{imu} is a residual threshold and w_e, w_p, w_s weight the terms. When excitation drops below τ_ω or residuals spike, the soft weight $\lambda_{\text{imu}} s_k^{(\text{IMU})}$ reduces the influence of unreliable inertial constraints.

Wheel slip consistency. Wheel odometry provides high-rate velocity measurements but is susceptible to slip and kinematic model violation [12], [48]. The scheduler compares wheel-derived increments $(\Delta\boldsymbol{p}^w, \Delta\psi^w)$ against inertial and visual predictions $(\Delta\boldsymbol{p}^I, \Delta\psi^I)$ within a short temporal window:

$$D_k^{(\text{Wheel})} = w_v \min\Big(1, \tfrac{\|\Delta\boldsymbol{p}^w - \Delta\boldsymbol{p}^I\|}{\tau_v}\Big) + w_\psi \min\Big(1, \tfrac{|\Delta\psi^w - \Delta\psi^I|}{\tau_\psi}\Big), \tag{22}$$

where τ_v, τ_ψ are translational and rotational deviation thresholds, and w_v, w_ψ weight the terms. Large discrepancies or low motion diversity inflate the wheel-factor covariance or set $s_k^{(\text{Wheel})} = 0$, limiting slip-induced drift while retaining reliable wheel constraints.

GNSS integrity checks. GNSS measurements suppress drift but can be corrupted by multipath or satellite blockage [14], [16], [26]. The scheduler examines fix quality q_{fix}, covariance trace $\text{tr}(\boldsymbol{\Sigma}_g)$, and innovation consistency against the local prediction $\|\boldsymbol{\nu}_g\|^2_{\boldsymbol{\Sigma}_g^{-1}}$:

$$D_k^{(\text{GNSS})} = w_q(1-q_{\text{fix}}) + w_\Sigma \min\Big(1, \tfrac{\text{tr}(\boldsymbol{\Sigma}_g)}{\tau_\Sigma}\Big) + w_\nu \min\Big(1, \tfrac{\|\boldsymbol{\nu}_g\|^2_{\boldsymbol{\Sigma}_g^{-1}}}{\tau_\nu}\Big), \tag{23}$$

where τ_Σ, τ_ν are covariance and innovation thresholds, and w_q, w_Σ, w_ν weight the terms. Failed integrity indicators exclude the GNSS factor through $s_k^{(\text{GNSS})}$, protecting the locally consistent trajectory from erroneous global updates. Thus, reliability control remains within a single optimization problem rather than switching among subsystems [18].

E. Online Spatiotemporal Calibration

Online Spatiotemporal Calibration (OSC) refines the LiDAR–IMU temporal offset and rotation extrinsic online under sufficient excitation. The calibration bundle is $\boldsymbol{\vartheta}_{\text{cal}} = \{\Delta t_{LI}, \mathbf{R}_{IL}\}$, with $\boldsymbol{t}_{IL}$ held fixed. OSC runs two lightweight workers in parallel: a temporal worker estimates Δt_{LI} from IMU–LiDAR motion alignment when excitation and turning guards are satisfied, while an extrinsic worker refines $\mathbf{R}_{IL}$ from scan-to-scan and inertial rotation consistency under multi-axis excitation and adequate scan quality. Candidate updates are admitted only after residual, confidence, and short-history consensus checks; accepted values are then injected into timestamp association, preintegration, and LiDAR deskewing while preserving the current world LiDAR pose.

1) Temporal Calibration

The temporal worker estimates Δt_{LI} on long asynchronous windows using frontend LiDAR odometry. LiDAR-frame angular velocity and LiDAR-derived acceleration surrogates are computed from consecutive front-end LiDAR odometry poses and interpolated using cubic Hermite splines, providing continuous motion cues for temporal alignment with IMU measurements. We use δ as the LiDAR-trajectory search shift: an IMU sample at t_i is matched to LiDAR motion at $t_i + \delta$, while the runtime offset is defined by

$$t_{\text{imu}} = t_{\text{lidar}} + \Delta t_{LI}, \qquad \Delta t_{LI} = -\delta. \tag{24}$$

A coarse candidate is initialized by maximizing the cross-correlation between IMU and shifted LiDAR motion norms,

$$\hat{\delta}_{\text{coarse}} = \arg \max_{\delta \in [-\Delta t_{\max}, \Delta t_{\max}]} \mathscr{C}\big(\|\boldsymbol{\omega}^I(t)\|,\ \|\boldsymbol{\omega}^L(t+\delta)\|\big), \tag{25}$$

optionally augmented by correlation between IMU specific-force norms and LiDAR-derived acceleration surrogate norms. The estimate is refined by minimizing a robust alignment cost on matched pairs,

$$\mathscr{J}_{LI}(\delta) = \frac{1}{N}\sum_i \rho_{\text{H}}\big(\|\boldsymbol{\omega}_i^I - \mathbf{R}_{IL}\boldsymbol{\omega}^L(t_i+\delta) - \boldsymbol{b}_g\|\big) + \lambda_a \rho_{\text{H}}\big(\|s\boldsymbol{a}_i^I - \mathbf{R}_{IL}\tilde{\mathbf{a}}^L(t_i+\delta) - \boldsymbol{b}_a\|\big), \tag{26}$$

where ρ_{H} is a Huber loss and $(\mathbf{R}_{IL}, \boldsymbol{b}_g, \boldsymbol{b}_a, s)$ are estimated jointly when the extrinsic is not yet locked; otherwise $\mathbf{R}_{IL}$ is fixed and only δ and gyro bias are refined. Accepted candidates must exceed confidence and excitation thresholds, improve the cost relative to a zero-offset baseline, and remain stable over a short candidate history; updates are suppressed during sharp turning. Repeated stable commits freeze Δt_{LI} for the runtime pipeline.

2) LiDAR–IMU Extrinsic Calibration

The extrinsic worker refines $\mathbf{R}_{IL}$ from motion-consistency constraints between inertial preintegration and a dedicated scan-to-scan LiDAR odometry branch that is isolated from the main SLAM frontend. For a motion pair over $[i, j]$, gyro integration gives

$$\mathbf{R}_{ij}^I = \prod_{k=i}^{j-1} \text{Exp}\big((\boldsymbol{\omega}_k - \boldsymbol{b}_g)\Delta t_k\big), \tag{27}$$

while calibration-branch scan-to-scan registration accumulates $\mathbf{R}_{ij}^L$. Under

$$\mathbf{R}_{ij}^I = \mathbf{R}_{IL}\mathbf{R}_{ij}^L\mathbf{R}_{IL}^\top, \tag{28}$$

the rotation vectors satisfy the linearized constraint

$$\boldsymbol{\phi}_{ij}^I \approx \mathbf{R}_{IL}\boldsymbol{\phi}_{ij}^L, \qquad \boldsymbol{\phi}_{ij}^I = \text{Log}(\mathbf{R}_{ij}^I), \quad \boldsymbol{\phi}_{ij}^L = \text{Log}(\mathbf{R}_{ij}^L). \tag{29}$$

Each pair is weighted by w_{ij}, which increases with scan-to-scan alignment quality. Calibration is admitted only when multi-axis excitation is sufficient: with $\boldsymbol{a}_k = \boldsymbol{\phi}_k^L/\|\boldsymbol{\phi}_k^L\|$ and $\mathbf{M} = \sum_k w_k \boldsymbol{a}_k \boldsymbol{a}_k^\top$, we require enough valid pairs, adequate accumulated rotation, and a lower bound on λ_1/λ_3 from the eigendecomposition of $\mathbf{M}$ to avoid single-axis degeneracy.

An initial rotation is obtained from weighted angular-velocity alignment,

$$\min_{\mathbf{R}_{IL}, \boldsymbol{b}_g} \sum_k w_k \big\|\mathbf{R}_{IL}\bar{\boldsymbol{\omega}}_k^L + \boldsymbol{b}_g - \bar{\boldsymbol{\omega}}_k^I\big\|^2, \tag{30}$$

which admits a closed-form Procrustes solution via SVD on the centered cross-covariance. If excitation or residual checks fail, a rotation-vector fallback minimizes $\sum_k w_k \|\mathbf{R}_{IL}\boldsymbol{\phi}_k^L - \boldsymbol{\phi}_k^I\|^2$ with the same SVD structure. The estimate is further refined by

$$\mathbf{R}_{IL} = \text{Exp}(\delta\boldsymbol{\theta})\mathbf{R}_0, \qquad \min_{\delta\boldsymbol{\theta}, \boldsymbol{b}_g} \sum_k \|e_k^R\|^2 + \sum_k \|e_k^\omega\|^2, \tag{31}$$

where e_k^R and e_k^ω are weighted rotation-vector and angular-velocity residuals around $\mathbf{R}_0$. Candidates passing mean/max residual, inlier-ratio, and excitation tests enter a consensus history; $\mathbf{R}_{IL}$ is locked once repeated agreement is reached across successful solves.

To avoid map discontinuity at lock time, the updated extrinsic is applied while preserving the current world LiDAR pose,

$$\mathbf{T}_{WI}^{new} = \mathbf{T}_{WI}^{old}\mathbf{T}_{IL}^{old}(\mathbf{T}_{IL}^{new})^{-1}, \qquad \mathbf{T}_{IL}^{new} = \{\mathbf{R}_{IL}^{\text{locked}}, \boldsymbol{t}_{IL}^{old}\}, \tag{32}$$

so that $\mathbf{T}_{WL}^{new} = \mathbf{T}_{WL}^{old}$ and the existing voxel map remains consistent with incoming scans.

F. Localization and Mapping Refinements

Several implementation refinements improve repeatability and mapping quality without changing the estimator:

Robust localization. Before residual construction, image/depth timestamps, LiDAR point times, intensity fields, trajectory frames, and ground-truth associations are aligned. Factors are restricted to physically supported state directions; confirmed low-excitation stops use a ZUPT-inspired branch [49], wheel factors are validated by IMU–wheel consistency, and LVIO feature frames are synchronized with the LiDAR window before reprojection.

Geometric and colorized mapping. State estimation is accompanied by a hybrid local map: a bounded voxel-hash map supports nearest-neighbor search, and a sliding downsampled map provides geometric support and plane-quality

evidence [24], [50]. Deskewed scans are transformed into the world frame and colorized through temporally aligned RGB projection for visualization. The mapping interface also supports LiDAR-guided 3D Gaussian Splatting [51]. Ultra-Fusion poses, aligned RGB images, and colorized LiDAR points provide metric anchors, sparse depth supervision, and appearance constraints for incremental Gaussian construction without replacing the odometry frontend.

IV. M3DGR Benchmark Dataset

M3DGR is designed for controlled evaluation of multi-sensor SLAM under degradation, calibration uncertainty, and diverse motion regimes. Its construction emphasizes broad modality coverage, accurate synchronization and calibration, and scenarios targeting representative transportation failure modes.

Relation to prior datasets. An earlier IROS version of M3DGR introduced the real-world degradation dataset and evaluated 40 representative SLAM systems [18]. This paper extends that benchmark in two ways: it adds simulation trajectories for controlled perturbation analysis, especially for LiDAR degeneracy and spatiotemporal miscalibration, and expands the comparison to more than 60 systems. The expanded benchmark is used both as a dataset contribution and as a stress-test protocol for configurable multi-sensor fusion systems.

A. Real-World Acquisition

Figure 4 summarizes the M3DGR acquisition setting, including benchmark scale, scenario taxonomy, representative trajectories, and both real and simulated sensing platforms. The real ground robot records RGB-D imagery, LiDAR point clouds, wheel encoder odometry, and raw GNSS measurements. LiDAR streams are transmitted through Ethernet, while the remaining sensors use USB 3.2; all data are logged on an Intel NUC with a high-speed NVMe SSD for long-duration synchronized acquisition.

As shown in the platform panel of Figure 4, the robot uses a differential-drive base with two driven wheels, which naturally provides wheel-odometry measurements. Visual sensing is provided by an RGB-D–IMU device and an omnidirectional camera. Two Livox solid-state LiDARs provide complementary 3D observations with non-repetitive scanning patterns, while GNSS/RTK receivers mounted on the top layer record satellite measurements for outdoor evaluation. For reference trajectories, we use an RTK receiver in outdoor sequences and a motion-capture system in indoor sequences.

The real-world part of M3DGR is organized around sensing failures that commonly affect ground mobility: appearance degradation for cameras, weak geometric constraints for LiDAR, unreliable wheel motion under slip or rough terrain, and intermittent GNSS availability. This design keeps the platform close to practical ITS deployment while making each dominant failure source observable during evaluation.

All sensor topics are recorded as ROS bags with a unified timestamping mechanism. In addition, several devices provide internal hardware-level synchronization to further reduce inter-sensor skew. Ground truth is provided by motion capture indoors, RTK GNSS outdoors, and ArUco-based start–end alignment for drift evaluation under degeneracy or GNSS outage, so both absolute trajectory accuracy and accumulated loop drift can be assessed.

B. M3DGR Sim Acquisition

The M3DGR Sim trajectories are recorded in NVIDIA Isaac Sim using a differential-drive mobile robot modeled after Nova Carter. As shown in Figure 4, the simulated platform provides a compact multi-sensor layout with body-mounted IMU, LiDAR, and stereo camera streams, while wheel-encoder measurements are generated from the two driven wheels.

For each scene, a closed-loop collection pipeline first constructs a 2D occupancy grid from multi-slice LiDAR projections in the 3D environment, then plans collision-free routes with obstacle-inflated A* search, simplifies them with Ramer–Douglas–Peucker, and tracks them using pure pursuit control. During autonomous motion, the simulated IMU, wheel encoder, LiDAR, and stereo camera share the same simulation-time clock, are published through the ROS 1 bridge, and are logged as synchronized bags together with simulator-derived sensor extrinsics as ground-truth calibration.

The simulated sequences complement real data by providing repeatable routes and exact calibration references. They are mainly used for controlled LiDAR-degeneracy and spatiotemporal-perturbation studies, where the same scene and motion can be replayed while changing timing offsets, extrinsic errors, or geometric structure.

C. Sequence Design

The sequences include routine operation and deliberately challenging scenes, enabling analysis of both average accuracy and modality-specific failure modes. Real-world sequences cover visual degradation, LiDAR degeneracy, wheel slip, GNSS outage, and standard short- and long-term operation, while M3DGR Sim adds Wild, Warehouse, and Tunnel trajectories for repeatable LiDAR-degeneracy and spatiotemporal-perturbation studies.

Each sequence group targets a different robustness question. Visual challenge sequences test whether localization remains stable under low light, illumination changes, dynamics, or occlusion; LiDAR-degenerate sequences emphasize corridor-like and elevator-transition geometry; wheel-slippage sequences expose biased planar motion constraints; GNSS-denial sequences test continuity without global updates; and standard routes provide nominal short- and long-duration baselines. Figure 4 visualizes representative trajectories by scenario; detailed sequence definitions, ground-truth sources, and aggregate statistics are provided in the supplementary material.

V. Experimental Evaluation

Evaluation is organized around configuration-wise accuracy, module causality, degradation robustness, calibration robustness, long-horizon/high-speed operation, and cross-platform validation.

- **Q1. Overall Benchmarking:** Does the Ultra-Fusion system with diverse configurations (WIO/VIO/LIO/LVIO with optional augmentation when applicable) achieve competitive performance against corresponding baselines?

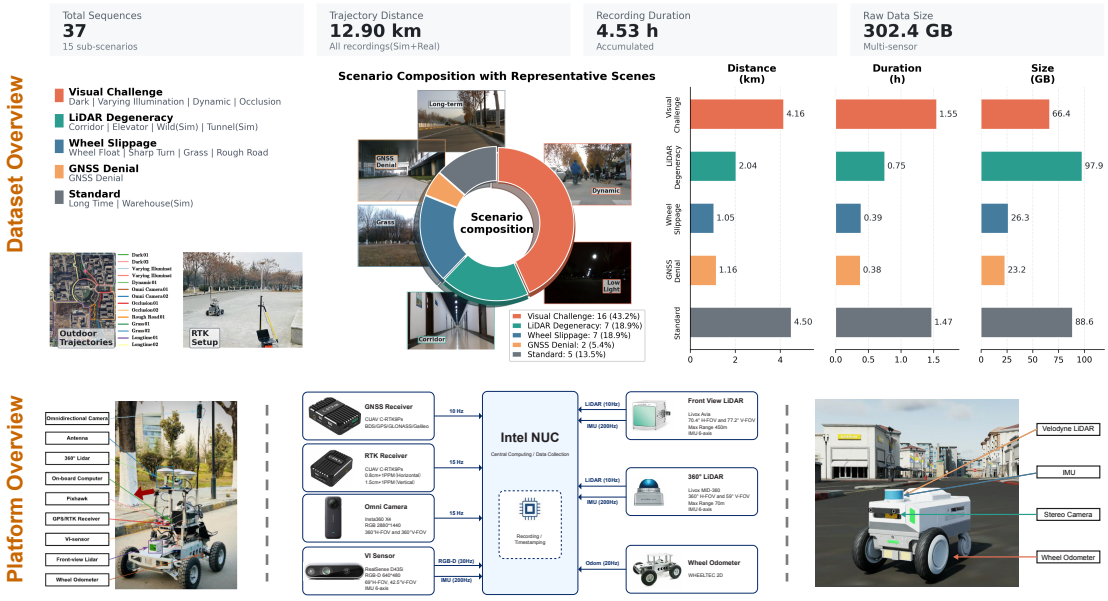

Fig. 4. Overview of the M3DGR benchmark and acquisition platforms. The upper panel summarizes the dataset scale, scenario composition, representative outdoor trajectories, and distance–duration–storage statistics across visual challenge, LiDAR degeneracy, wheel slippage, GNSS-denial, standard, and simulation sequences. The lower panel shows the real ground-robot platform and the M3DGR Sim robot, including mechanical layout, sensor placement, and data links among the RGB-D–IMU camera, omnidirectional camera, dual LiDARs, GNSS/RTK receivers, wheel odometer, Pixhawk, onboard computer, and simulated IMU–wheel–LiDAR–stereo setup.

- **Q2. Degradation Robustness:** Does the degradation-aware tightly-coupled design improve robustness under modality-specific failure modes?
- **Q3. Spatiotemporal Calibration:** Does online LiDAR–IMU temporal and extrinsic calibration remain effective under injected delay and rotation perturbations?
- **Q4. Long-Term and High-Speed Operation:** Does Ultra-Fusion remain robust and stable over long-duration and high-speed trajectories?
- **Q5. Cross-Platform Validation:** Does Ultra-Fusion remain effective on representative heterogeneous robotic platforms?

Experiments use M3DGR [18] for degradation and controlled perturbation on a wheeled ground robot, M2DGR-Plus [7] for campus-scale wheeled routes, KAIST [42] for city-scale driving up to 96.9 km/h, GrandTour [44] for quadruped mobility, and MARS-LVIG [43] for low-altitude UAV trajectories along airport infrastructure.

A. Comprehensive Benchmarking

Table II compares Ultra-Fusion with representative systems on ten M3DGR sequences, emphasizing accuracy and robustness under heterogeneous degradation. M2DGR-Plus results are reported in Table III.

Baselines and evaluation metrics. Baselines are grouped by compatible sensor availability: wheel/GNSS references, visual systems without LiDAR [7], [8], [12], [19]–[21], [26], [52]–[55], LiDAR-only or LiDAR–inertial systems [23], [24], [29]–[31], [50], [56]–[86], and LiDAR–visual systems [3], [11], [15], [18], [25], [32], [87], [88]. Each method uses its supported streams, and each Ultra-Fusion variant uses only the modalities indicated by its mode name. Accuracy is reported as EVO-aligned ATE RMSE. A run is marked by ✗if initialization, tracking, execution, or temporal alignment fails; failed runs are penalized in aggregate ATE as specified in the table note. Repository links and licenses are summarized in the supplementary material. **We will release the implementation and evaluation scripts upon paper acceptance.**

Overall performance. Table II reveals modality-dependent failure modes. Visual pipelines degrade under poor correspondence [19], [53], [54], whereas LiDAR-based systems tolerate appearance changes but may fail under geometric degeneracy [15], [23], [88]. Average rank is therefore more informative than isolated best-case accuracy.

Ultra-Fusion achieves favorable ranks across sensor groups. WIO improves over raw wheel odometry and Ground-Fusion WIO, while VWIO, LWIO, and LVWIO lead the visual-based, LiDAR-only, and LiDAR–visual groups. The gains are consistent with observability-aware initialization and factor-level fusion: wheel/GNSS availability changes active factors without altering the estimator, and LiDAR residuals remain optimized in the same window rather than injected as an external odometry prior. On M3DGR, lower ATE under visual, LiDAR-degenerate, wheel-slip, and GNSS-denial sequences reduces pose errors in tunnel transit, degraded-camera driving, and mixed indoor–outdoor mobility. On M2DGR-Plus (Table III), Ultra-Fusion achieves the lowest average drift rate and RMSE (0.59% / 0.24 m), compared with 2.32% / 1.48 m for FAST-LIVO2 [15] and 1.71% / 0.75 m for Ground-Fusion [7].

B. Module Ablations

Reliability Scheduling Figure 5 compares Ultra-Fusion with and without each FRS component while keeping the sensor stream active. LiDAR FRS is evaluated in LVWIO, whereas visual and wheel FRS are evaluated in VWIO to avoid masking by dominant LiDAR constraints. Enabling FRS reduces mean ATE by 0.45 m for LiDAR (75.3%), 1.60 m for vision (36.2%), and 1.56 m for wheel odometry (41.3%). GNSS gating yields

TABLE II. ATE RMSE (M) COMPARISON OF REPRESENTATIVE SLAM SYSTEMS ON M3DGR SEQUENCES.

Method/Scenario	Summary	Visual Challenge				LiDAR Degeneracy		Wheel Slippage			GNSS Denial
	Avg. Rank[1]/ATE	Dynamic01	Varying-illu01	Dark01	Occlusion01	Corridor01	Elevator01	Wheel-float01	Sha-turn01	Grass01	GNSS-denial01
Raw Wheel Odom	2.5/35.6	8.60	8.21	**5.48**	6.90	74.32	*66.94*	*1.18*	*7.44*	26.95	✗
Ground-Fusion WIO [7]	*2.0/33.68*	*2.32*	*2.36*	*5.52*	*2.04*	*72.61*	*66.94*	2.20	*7.44*	25.34	✗
GNSS SPP	2.7/106.98	✗	✗	7.69	✗	✗	✗	✗	✗	**0.48**	**11.61**
Ultra-Fusion (WIO), 2026	**1.3/26.99**	**0.55**	**0.48**	5.59	**0.81**	**24.67**	**64.55**	**0.49**	**2.06**	*20.65*	✗
ORB-SLAM2 [52], 2017	7/76.79	**0.14**	✗	✗	✗	6.41	8.09	1.72	1.54	✗	✗
VINS-Mono [21], 2018	6.4/26.7	0.43	2.70	7.91	✗	9.82	62.80	0.46	0.36	2.17	30.36
VINS-RGBD [8], 2019	5.7/49.21	0.20	1.86	✗	✗	5.62	✗	0.28	0.35	7.52	26.31
TartanVO [53], 2021	8.5/62.56	2.37	2.17	12.37	✗	✗	✗	1.93	2.09	4.68	✗
ORB-SLAM3 [19], 2021	9.8/150	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
VINS-GPS-Wheel [55], 2021	6.7/25.53	1.18	1.32	15.55	✗	5.55	43.48	0.86	2.00	18.47	16.89
DM-VIO [20], 2022	7.7/63.22	2.25	2.27	4.08	✗	12.20	**2.54**	✗	8.90	✗	✗
GVINS [26], 2022	5.7/61.45	0.26	1.25	✗	✗	9.42	*2.89*	0.27	0.40	✗	✗
VIW-Fusion [12], 2022	5.5/27.99	0.62	1.02	**0.77**	✗	5.58	16.68	0.77	2.44	2.91	99.06
Ground-Fusion [7], 2024	4/7.52	*0.19*	*0.59*	*1.10*	**1.21**	26.25	29.93	0.29	1.16	*1.33*	13.19
MASt3R-SLAM [54], 2025	6.8/90.14	0.35	**0.30**	✗	✗	✗	✗	0.31	0.40	✗	✗
Ultra-Fusion (VIO), 2026	*3.2/18*	0.23	1.20	4.80	✗	*5.30*	6.50	*0.20*	**0.29**	4.60	*6.90*
Ultra-Fusion (VWIO), 2026	**2.1/2.23**	0.20	0.66	1.20	*1.30*	**4.50**	6.70	**0.18**	*0.33*	**1.05**	**6.15**
A-LOAM [56], 2018	13.6/13.22	0.15	0.16	6.36	0.19	66.66	48.37	0.29	0.26	1.29	8.46
LeGO-LOAM [59], 2018	19.8/49.01	7.92	✗	13.40	6.28	19.65	✗	5.89	8.40	32.67	95.91
LIO-mapping [76], 2019	18.3/66.53	2.03	1.95	✗	2.46	✗	✗	1.05	1.46	56.3	✗
LIO-SAM [22], 2020	16.7/38.99	5.10	2.24	✗	1.31	36.15	✗	0.73	0.63	0.56	43.21
LINS [72], 2020	20.2/58.05	10.18	3.40	13.25	5.07	✗	✗	5.03	5.04	88.55	✗
LOAM-Livox [57], 2020	17.9/30.01	2.88	3.24	2.55	2.42	43.91	87.52	1.47	1.78	4.36	✗
LiLi-OM [67], 2021	17.3/47.93	1.49	0.19	✗	7.00	✗	✗	0.35	2.08	2.22	15.96
LIO-Livox [70], 2021	14.7/32.04	0.18	0.72	0.30	0.47	✗	✗	0.35	11.25	0.54	6.63
Faster-LIO [24], 2022	11.4/31.17	0.12	0.13	0.17	0.11	✗	✗	2.19	2.84	0.5	5.60
IESKF-LIO [75], 2022	6.9/16.59	0.14	0.14	0.15	0.13	14.38	✗	0.17	0.16	0.54	*0.08*
VoxelMap [50], 2022	10.9/3.07	0.89	0.76	4.93	0.91	1.08	19.22	0.99	1.11	0.78	**0.07**
FAST-LIO2 [23], 2022	9.7/31.49	0.13	0.11	0.24	0.11	✗	✗	0.16	0.18	0.51	13.46
CTLO [58], 2023	6.5/5.79	*0.10*	0.12	0.15	0.14	3.29	52.31	0.18	0.15	0.54	0.88
Point-LIO [29], 2023	10.8/17.96	0.14	0.14	0.29	0.15	9.58	✗	0.19	0.20	0.52	18.39
LOG-LIO [68], 2023	10.6/32.92	0.13	0.12	0.99	0.14	✗	✗	0.18	**0.07**	0.53	27.03
CT-LIO [69], 2023	6.3/1.00	0.12	0.12	0.18	0.13	3.56	2.39	**0.13**	*0.10*	0.55	2.71
DLIO [71], 2023	8/9.24	0.12	*0.10*	0.16	0.15	40.46	44.70	0.18	0.17	0.51	5.80
HM-LIO [74], 2023	6.8/15.48	0.12	0.15	0.17	0.14	3.23	✗	*0.14*	0.20	0.53	0.09
KISS-ICP [60], 2023	10.2/30.29	0.15	0.15	0.15	0.17	✗	✗	0.25	0.22	*0.53*	1.3
SLICT [30], 2023	9.6/30.24	0.14	0.14	0.17	0.16	✗	✗	0.25	0.20	*0.53*	0.81
MM-LINS [73], 2024	16.5/24.63	2.84	2.79	0.27	1.95	✗	74.31	2.25	2.83	1.29	7.73
SLICT2 [31], 2024	8.2/15.44	0.14	0.14	0.17	0.16	0.87	✗	0.24	0.20	*0.53*	1.91
PIN-SLAM [61], 2024	8.4/30.15	0.13	0.14	**0.05**	0.16	✗	✗	0.25	0.18	*0.53*	0.10
I2EKF-LO [62], 2024	6.9/3.49	2.37	0.13	0.16	0.15	1.31	29.7	0.24	0.20	*0.53*	*0.08*
LTAOM [80], 2024	8/15.20	0.16	0.17	0.18	0.18	0.19	✗	0.26	0.25	0.54	*0.08*
LOG-LIO2 [77], 2024	9.7/30.18	0.15	0.15	0.18	0.17	✗	✗	0.25	0.22	0.54	0.09
Eq-LIO [81], 2024	6.7/15.19	0.14	0.14	0.18	0.16	0.20	✗	0.24	0.20	*0.53*	**0.07**
Traj-LO [63], 2024	7.1/15.36	0.13	0.13	0.15	0.15	0.87	✗	0.24	0.19	**0.52**	1.18
VoxelMap++ [83], 2024	12.3/30.55	0.21	0.18	0.22	0.22	✗	✗	0.24	0.20	1.23	3.03
DMSA-SLAM [84], 2024	10.8/45.2	0.13	0.13	✗	0.15	✗	✗	0.24	0.20	0.56	0.58
Adaptive-LIO [79], 2025	*3.6/0.23*	*0.10*	*0.10*	0.17	*0.10*	0.26	0.57	0.18	0.13	0.54	0.10
GLO [65], 2025	11.7/31.24	0.15	0.15	0.23	0.16	✗	✗	0.23	0.20	0.63	10.62
LIGO [16], 2025	9.1/6.62	*0.10*	0.12	0.25	0.16	9.55	43.34	0.19	0.17	0.55	11.75
CTE-MLO [64], 2025	8.2/30.16	0.13	0.13	0.15	0.15	✗	✗	0.25	0.20	*0.53*	**0.07**
RKO-LIO [82], 2025	8.2/15.41	0.15	0.14	0.17	0.16	2.03	✗	0.25	0.21	0.54	0.48
II-NVM [85], 2025	10/4.17	0.14	0.14	3.77	0.16	19.00	14.88	0.24	0.20	0.56	2.62
Surfel-LIO [86], 2025	9.2/15.51	0.16	0.16	0.18	0.18	2.87	✗	0.26	0.25	0.55	0.44
GenZ-ICP [66], 2025	6.4/0.41	0.14	0.14	0.15	0.16	0.32	1.61	0.25	0.21	*0.53*	0.63
Voxel-SLAM [89], 2026	4.6/0.55	**0.09**	**0.09**	0.18	**0.09**	1.41	1.45	0.16	*0.10*	0.54	1.39
Ultra-Fusion (LIO), 2026	3.8/0.19	0.12	0.14	*0.08*	0.15	*0.17*	**0.25**	*0.14*	0.20	0.54	**0.07**
Ultra-Fusion (LWIO), 2026	**3.5/0.17**	0.12	0.12	*0.08*	0.14	**0.03**	*0.29*	*0.14*	0.20	0.54	*0.08*
LVI-SAM [11], 2021	6.4/22.88	0.85	136.03	4.23	30.85	7.06	28.44	0.64	0.49	7.63	12.56
R2LIVE [87], 2021	3.2/30.34	**0.11**	**0.11**	0.13	**0.10**	✗	✗	**0.09**	**0.19**	1.33	1.36
R3LIVE [3], 2022	6.9/33.8	8.76	4.24	1.12	9.00	6.07	✗	1.07	6.00	1.69	✗
FAST-LIVO [25], 2022	7.3/63.06	✗	8.95	✗	9.49	7.96	✗	0.78	1.92	1.50	✗
Coco-LIC [32], 2023	5.3/16.61	1.77	0.97	0.54	1.66	6.98	✗	0.64	1.80	1.21	0.54
SR-LIVO [88], 2024	6.5/69.59	1.23	0.28	*0.09*	1.31	✗	✗	0.86	✗	✗	92.14
FAST-LIVO2 [15], 2024	4.5/16.57	0.44	0.28	0.17	0.33	3.35	✗	0.51	0.81	9.71	0.09
Ground-Fusion++ [18],2025	3.5/1.1	0.13	0.14	0.17	0.16	5.69	2.48	0.20	*0.22*	1.39	0.40
Ultra-Fusion (LVIO), 2026	*1.8/0.22*	*0.12*	*0.13*	**0.08**	0.15	*0.47*	*0.28*	*0.14*	**0.19**	**0.53**	*0.08*
Ultra-Fusion (LVWIO), 2026	**1.4/0.15**	*0.12*	**0.11**	**0.08**	*0.13*	**0.02**	**0.06**	*0.14*	**0.19**	*0.54*	**0.07**

[1] In the Avg. Rank/ATE summary, failed runs are counted as 150 m.

TABLE III. COMPARISON OF ATE RMSE (M) / DRIFT RATE ON THE M2DGR-PLUS DATASET [7].

Method/Scenario	Anomaly	Switch	Tree	Building1	Building2	Bridge2	Parking1	Parking2	Street1	Street2
FAST-LIVO [25]	✗	✗	✗	✗	✗	✗	1.06 / 3.80%	✗	0.38 / 2.53%	✗
FAST-LIVO2 [15]	*0.10 / 1.36%*	*1.60 / 1.86%*	2.70 / 3.73%	1.62 / 3.51%	✗	4.99 / 2.34%	0.81 / 2.90%	0.54 / 1.27%	*0.35 / 2.33%*	0.57 / 1.56%
Ground-Fusion [7]	0.29 / 3.96%	1.80 / 2.09%	✗	*0.60 / 1.29%*	*0.29 / 1.04%*	*2.37 / 1.11%*	*0.48 / 1.73%*	*0.19 / 0.44%*	0.47 / 3.13%	*0.23 / 0.63%*
Ground-Fusion++ [18]	✗	✗	*3.40 / 4.70%*	✗	1.68 / 6.00%	✗	✗	✗	✗	✗
Ultra-Fusion (LVWIO)	**0.09 / 1.23%**	**0.23 / 0.27%**	**0.16 / 0.21%**	**0.32 / 0.69%**	**0.26 / 0.93%**	**0.81 / 0.38%**	**0.10 / 0.36%**	**0.09 / 0.21%**	**0.20 / 1.33%**	**0.12 / 0.33%**

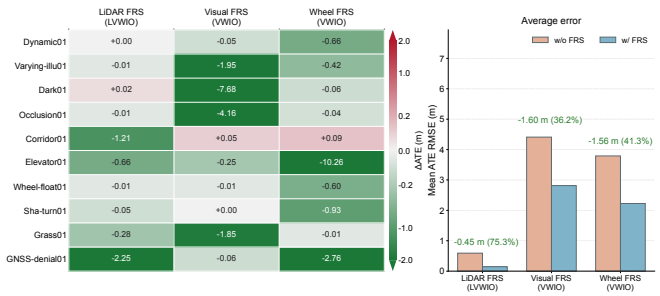

Fig. 5. Ablation of Factor-Wise Reliability Scheduling (FRS) on M3DGR.

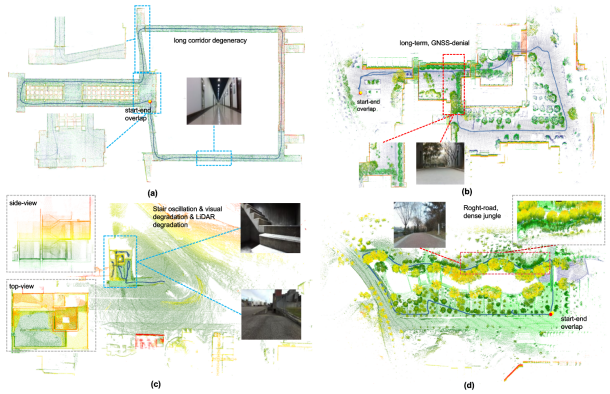

Fig. 6. Qualitative robustness under representative sensor-degradation scenarios. Estimated trajectories (blue) and colored point-cloud maps are shown for four stress sequences; red markers denote start–end overlap. (a) M3DGR Corridor01: long-corridor LiDAR degeneracy with weak geometric constraints. (b) M3DGR GNSS-denial01: prolonged GNSS outage with localized visual degradation. (c) GrandTour ARC-2: quadruped stair traversal with body oscillation and concurrent visual/LiDAR degradation. (d) M3DGR Z-Rough-Road01: long-distance navigation in dense vegetation with high visual ambiguity.

negligible change on Dark01 (0.0628 m), improves Grass01 from 0.618 m to 0.539 m, and reduces GNSS-denial01 from 2.77 m to 1.79 m, indicating selective rejection of unreliable satellite updates.

Initialization Observability-Aware Initialization is ablated by disabling the adaptive bootstrap path and retaining only static IMU/gravity initialization, with pose output suppressed until LiDAR-odometry-aided MAP initialization succeeds (Ultra-Fusion with adaptive initialization). Table IV summarizes 18 sequences with complete 20 s outputs across LVIG, KAIST, M3DGR, M2DGR-Plus, and GrandTour. The full system achieves mean initialization latency of 0.153 s, median latency of 0.150 s, and mean 20 s ATE of 0.483 m, with the fastest initialization in 15/18 sequences and the lowest 20 s ATE in 11/18. Disabling adaptive initialization raises mean latency to 4.642 s and mean 20 s ATE to 16.808 m.

TABLE IV. INITIALIZATION LATENCY AND EARLY-WINDOW ACCURACY ON SEQUENCES WITH COMPLETE 20 S OUTPUTS FROM ALL COMPARED SEQUENCES.

Method	Mean init (s)	Median init (s)	Mean 20 s ATE (m)
Ground-Fusion++ [18]	2.118	1.058	85.217
FAST-LIVO2 [15]	0.913	0.671	28.687
FAST-LIVO [25]	1.436	1.192	73.754
Ultra-Fusion (w/o adaptive initialization)	4.642	2.097	16.808
Ultra-Fusion	**0.153**	**0.150**	**0.483**

Failed runs are assigned an initialization time of 20 s and ATE of 100 m.

C. Robustness under Sensor Degradation

The degradation study isolates modality-specific failure mechanisms on M3DGR and complements them with M2DGR-Plus trajectories where multiple degradations may co-occur [7].

Visual degradation: Visual degradation weakens feature association and photometric consistency; ORB-SLAM3 [19], TartanVO [53], and MASt3R-SLAM [54] are therefore unstable under darkness or occlusion. Ultra-Fusion admits visual factors only when feature support, spatial distribution, reprojection consistency, and feature-frame availability are sufficient, allowing LiDAR, inertial, and wheel factors to dominate when visual observability degrades. On long-range routes through dense vegetation, such as Z-Rough-Road01 in Fig. 6(d), heavy occlusion and visual ambiguity can deteriorate correspondence quality; reliability gating suppresses inconsistent visual constraints while complementary modalities preserve trajectory closure.

LiDAR degeneracy: LiDAR degeneracy stems from rank-deficient geometry. Corridor and elevator scenes reduce constraint diversity, causing FAST-LIO2 [23], FAST-LIVO2 [15], and SR-LIVO [88] to degrade or fail. Ultra-Fusion scores LiDAR factors by geometric conditioning and constraint diversity, enabling LIO/LWIO to down-weight ill-conditioned scans and LVWIO to regularize weak directions with complementary modalities. Fig. 6(a) shows Corridor01, where extended corridor degeneracy weakens observability yet Ultra-Fusion avoids performance collapse and closes the loop with negligible start–end error.

The optional intensity cue is evaluated separately. Intensity-consistency residuals are activated only with sufficient local support and serve as a complementary safeguard in geometrically weak regions, rather than replacing geometric registration. Table V shows that Ultra-Fusion keeps submeter accuracy on Wild scenes and substantially reduces drift in tunnel degeneracy, with the intensity cue improving the Wild sequences and matching geometry-only performance on Tunnel01. Fig. 7(a) shows the M3DGR Sim tunnel sequence Tunnel02, where LiDAR-only methods often fail in a geometrically degenerate segment; reliability scheduling down-weights ill-conditioned LiDAR factors while visual and inertial cues preserve continuity.

TABLE V. COMPARISON OF ATE RMSE (M) ON LIDAR-DEGENERATE ISAAC SIM SEQUENCES.

Method	Wild01	Wild02	Tunnel01	Tunnel02
R3LIVE [3]	308.03	210.97	✗	1310.17
FAST-LIVO [25]	5.31	309.44	1.05	24.64
FAST-LIVO2 [15]	5.94	12.96	*0.13*	14.42
Ultra-Fusion (w/o intensity)	*0.10*	*0.95*	**0.08**	**2.07**
Ultra-Fusion (w/ intensity)	**0.06**	**0.70**	**0.08**	*2.21*

Wheel slippage: Wheel slippage introduces systematic bias that may not be evident from residual magnitude. Raw wheel odometry and Ground-Fusion WIO degrade under slip, weak excitation, or long horizons. Ultra-Fusion (WIO) reduces average ATE from 35.6 m and 33.68 m to 26.99 m by restricting wheel constraints to their planar observable subspace and validating them with inertial, visual, or LiDAR motion consistency.

GNSS denial: GNSS denial tests whether global positioning is auxiliary or essential. Methods relying heavily on abso-

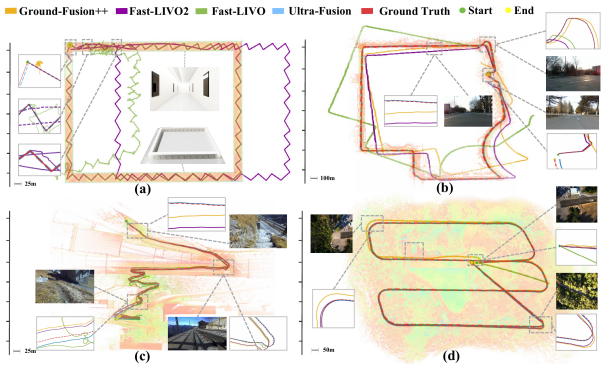

Fig. 7. Trajectory estimates on four ITS-relevant sequences: (a) M3DGR Sim Tunnel02, (b) M3DGR Longtime02, (c) GrandTour EIG-1 quadruped, and (d) MARS-LVIG HKAirport02 UAV.

lute updates, such as VINS-GPS-Wheel [55] and Ground-Fusion [7], drift when GNSS becomes unavailable. In Ultra-Fusion, GNSS is an integrity-checked factor, so local LiDAR, visual, inertial, and wheel constraints continue without reinitialization when GNSS degrades.

On Longtime02 (>30 min), enabling GNSS reduces ATE RMSE from 17.40 m to 8.45 m, indicating that integrity-checked global updates limit drift over extended wheeled patrol trajectories. Fig. 8(b)–(c) shows Longtime02 and GNSS-denial01, where local constraints continue through GNSS-denied segments. Fig. 6(b) further shows accurate convergence to the origin despite prolonged satellite outage and localized visual degradation.

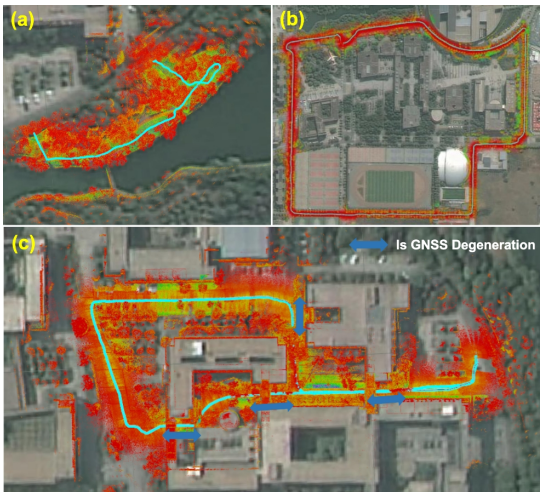

Fig. 8. Evaluation of GNSS augmentation and GNSS-denial robustness. (a) Grass01 trajectory overlaid on satellite imagery. (b) Longtime02 (>30 min) trajectory with a large-scale loop. (c) GNSS-denial01 sequence, where blue arrows indicate GNSS-denied segments.

Overall, robustness derives from sensor redundancy and factor-wise reliability scheduling: informative measurements are retained, while inconsistent or degenerate constraints are suppressed. Remaining failures correspond to configurations lacking observations for specific degrees of freedom, rather than forced compensation by low-confidence residuals.

D. Robustness under Spatiotemporal Miscalibration

Spatiotemporal robustness is evaluated by injecting LiDAR–IMU temporal delays and extrinsic rotation perturbations,

TABLE VI. RMSE (M) UNDER INJECTED IMU TIME OFFSETS ON WILD01.

Time offset	FAST-LIVO [25]	FAST-LIVO2 [15]	Ground-Fusion [7]	Ground-Fusion++ [18]	Ultra-Fusion (w/o OSC[1])	**Ultra-Fusion (full)**
0ms	8.6999	8.0406	3.4194	8.8026	0.0484	0.0319
+100ms	38.1436	10.4990	4.1732	8.9272	0.0718	0.0348
+200ms	77.0621	9.7028	10.0682	9.0952	0.4684	0.0344
+300ms	7.0270	6.5562	3.5356	9.2159	0.6919	0.0346
-100ms	118.8689	7.8137	3.1553	9.2141	0.0629	0.0356
-200ms	157.5297	10.8993	30.4176	8.6085	0.1304	0.0375
-300ms	16.7918	15.4956	3.5999	8.4933	1.0094	0.0403

[1] Without online LiDAR–IMU temporal calibration (OSC temporal thread).

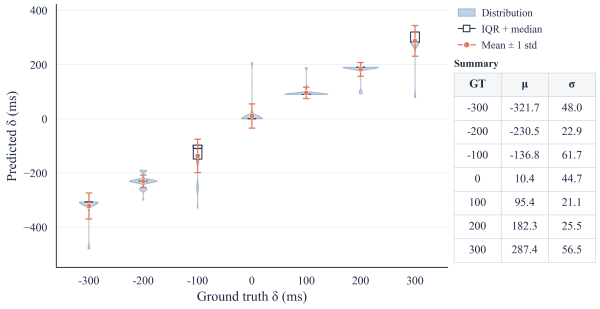

Fig. 9. Predicted temporal offsets δ on Wild01 under injected LiDAR-IMU timing perturbations.

corresponding to the two parallel OSC threads in Sec. III-E. Table VI and Fig. 9 test the temporal worker on the M3DGR Sim open-area sequence Wild01 with injected IMU delays. Ultra-Fusion degrades more gradually than the baselines and remains below one meter up to ± 200 ms. Disabling OSC increases error as delay grows, whereas the full system keeps sub-decimeter RMSE and the predicted δ values concentrate near the injected offsets. Table VII tests the extrinsic rotation worker on HILTI22 [90]. FAST-LIVO loses tracking on this fast-motion sequence even at 0° perturbation, so its large RMSE reflects trajectory failure rather than rotation sensitivity alone; FAST-LIVO2 and Ground-Fusion++ remain sensitive to injected rotation errors, whereas Ultra-Fusion maintains sub-meter RMSE over 0–10° perturbations with online extrinsic calibration enabled.

TABLE VII. RMSE (M) EVALUATION ON CORRIDOR LOWER GALLERY 2 FROM HILTI22 WITH EXTRINSIC ROTATION PERTURBATIONS

Rotation Error (°)	FAST-LIVO [25]	FAST-LIVO2 [15]	Ground-Fusion++ [18]	Ultra-Fusion (w/o extr. calib.[1])	**Ultra-Fusion (full)**
0.00	✗	0.15	2.93	0.12	**0.10**
1.00	✗	0.18	3.91	**0.09**	0.12
2.00	✗	0.18	3.94	0.19	**0.10**
3.00	✗	0.13	2.44	0.34	**0.10**
4.00	✗	0.26	5.52	0.23	**0.18**
5.00	✗	0.61	3.83	0.41	**0.11**
6.00	✗	0.81	3.38	0.47	**0.12**
7.00	✗	8.81	1.50	0.43	**0.14**
8.00	✗	145.11	2.75	0.44	**0.10**
9.00	✗	27.80	2.07	0.68	**0.12**
10.00	✗	940.37	3.81	0.75	**0.25**

[1] Without online LiDAR–IMU extrinsic calibration. FAST-LIVO loses tracking on HILTI22 at all perturbation levels.

E. Long-Term and High-Speed Operation

Long-duration M3DGR trajectories and city-scale high-speed KAIST urban-driving sequences evaluate prolonged operation, rapid motion, and accumulated calibration error. On Longtime01 and Longtime02, both exceeding 30 minutes, Ultra-Fusion obtains the lowest positioning errors (Table VIII), which limits drift accumulation during extended shuttle or patrol missions.

TABLE VIII. COMPARISON OF ATE RMSE (M) ON LONG-DURATION M3DGR SEQUENCES.

Methods	Longtime01	Longtime02
FAST-LIVO [25]	20.5	27.5
FAST-LIVO2 [15]	*5.13*	*8.4*
Ground-Fusion [7]	22.5	✗
Ground-Fusion++ [18]	7.5	15.9
Ultra-Fusion (LVWIO)	**4.3**	**2.8**

On the city-scale KAIST dataset, with speeds from 25.2 to 96.9 km/h, Ultra-Fusion (LVWIO) reduces average drift to 0.38% (Table IX). FAST-LIVO and FAST-LIVO2 exhibit large drift on these high-speed trajectories. Wheel–inertial motion support and LiDAR/visual drift correction remain complementary when optimized in a common window. Fig. 7(b) shows the Longtime02 trajectory over a 30+ minute route with repeated revisits to mapped regions.

F. Cross-Platform Validation

We evaluate whether the Ultra-Fusion framework transfers beyond the wheeled settings. We use GrandTour for legged robots and MARS-LVIG for aerial robots, covering body oscillation, shocks, large viewpoint changes, rapid motion, and weak structural priors.

Legged Robots: GrandTour [44] evaluates transfer to quadruped platforms relevant to last-mile delivery and facility inspection, using Relative Trajectory Error over 0.5-meter path segments. Body oscillation, attitude variation, shocks, and rapid viewpoint changes challenge registration and tracking. Table X shows that Ultra-Fusion obtains the lowest error on three of four sequences and remains competitive on the remaining one, so the estimator is not restricted to wheel-specific motion priors. Fig. 7(c) shows the GrandTour EIG-1 quadruped sequence on uneven terrain, where close alignment with ground truth is maintained despite slip-prone contacts and visual occlusion. Fig. 6(c) further reports ARC-2, where up–down stair motion induces platform oscillation and abrupt viewpoint changes that limit visual observability; Ultra-Fusion remains stable without accumulated drift.

Aerial Robots: For aerial validation, MARS-LVIG [43] provides UAV trajectories at 80–130 m altitude for corridor inspection, airport surroundings, and low-altitude urban mapping. Large viewpoint variation, rapid motion, and weak structural priors cause several baselines to degrade or fail. Table XI shows that Ultra-Fusion achieves low average error, supporting transfer beyond ground-robot motion statistics and yielding georeferenced trajectories suitable for infrastructure monitoring along transportation corridors. Fig. 7(d) shows the MARS-LVIG HKAirport02 UAV sequence at about 80 m altitude in an urban airport environment.

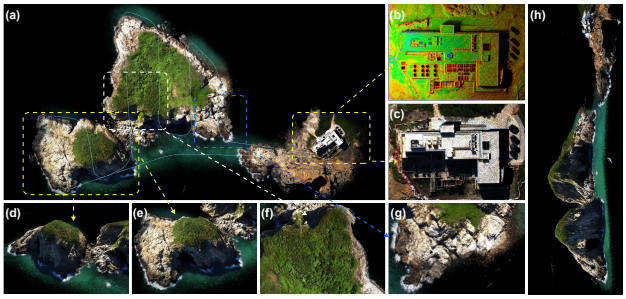

Fig. 10. Reconstruction map of the MARS-LVIG HKisland03 sequence. (a) Global top-view map with the estimated trajectory and highlighted regions. (b)–(c) Intensity-colored and RGB-colored details of the building area. (d)–(e) Enlarged views of the lower-left island. (f)–(g) Enlarged views of the central island. (h) Side-view reconstruction showing the 3D scene structure.

TABLE IX. ATE RMSE (M) / DRIFT RATE COMPARISON ON KAIST [42] DATASET. SEQUENCE HEADERS LIST TRAJECTORY LENGTH AND AVG. SPEED.

Method / Seq. Info.	Urban23 3379.7 m / 96.9 km/h	Urban25 2505.1 m / 91.4 km/h	Urban26 3987.8 m / 25.2 km/h	Urban29 3559.0 m / 29.1 km/h	Urban35 3187.7 m / 67.0 km/h
FAST-LIVO [25]	979.84 / 28.99%	714.29 / 28.51%	487.91 / 12.24%	822.62 / 23.11%	873.49 / 27.40%
FAST-LIVO2 [15]	979.40 / 28.98%	705.77 / 28.17%	497.03 / 12.46%	818.37 / 22.99%	918.62 / 28.82%
Ground-Fusion [7]	1825.70 / 54.02%	486.74 / 19.43%	173.52 / 4.35%	✗	703.14 / 22.06%
Ground-Fusion++ [18]	586.28 / 17.35%	✗	✗	✗	✗
VINS-GPS-Wheel [55]	✗	✗	✗	✗	✗
Raw Wheel Odom	284.60 / 8.42%	*14.63 / 0.58%*	*36.15 / 0.91%*	*19.42 / 0.55%*	4.44 / 0.14%
Ultra-Fusion (WIO)	*279.90 / 8.28%*	14.69 / 0.59%	36.56 / 0.92%	19.88 / 0.56%	*4.43 / 0.14%*
Ultra-Fusion (LVIO)	334.77 / 9.91%	441.68 / 17.63%	428 / 10.22%	525.25 / 14.76%	671.15 / 21.05%
Ultra-Fusion (LVWIO)	**12.38 / 0.37%**	**14.56 / 0.37%**	**32.50 / 0.58%**	**16.43 / 0.46%**	**4.12 / 0.13%**

TABLE X. COMPARISON OF RTE (CM) ON GRANDTOUR [44][1] ACROSS REPRESENTATIVE LEGGED-ROBOT SEQUENCES.

Method / Seq. Info.	SPX-2 urban / large-scale	SNOW-2 snowy / low-visibility	EIG-1 industrial / cluttered	ARC-2 debris / unstructured
Traj-LO [63]	1.11	1.28	1.12	17.36
DLO [91]	1.77	3.07	2.49	2.91
I2EKF-LO [62]	1.22	4.04	4.29	3.69
CT-LO [46]	1.55	1.45	1.53	4.23
CTE-MLO [64]	2.12	1.21	1.65	11.13
GenZ-ICP [66]	1.51	26.40	3.75	3.09
RESPLE-LO [92]	1.40	2.01	2.02	3.71
KISS-ICP [60]	3.28	35.78	7.76	13.71
Coco-LIC [32]	*0.44*	*0.41*	*0.40*	1.01
FAST-LIVO [25]	69.98	69.62	75.13	63.01
FAST-LIVO2 [15]	1.05	1.25	1.11	**0.70**
Ground-Fusion [7]	983.19	6.42	✗	30.07
Ground-Fusion++ [18]	1.82	1.90	3.96	2.83
PV-LIO [93]	1.14	1.32	1.24	1.05
DLIO [71]	1.22	1.51	1.26	1.19
Fast-LIMO [94]	1.15	1.16	1.10	2.49
Voxel-SLAM [89]	1.03	1.37	1.34	1.24
Ultra-Fusion (LVIO)	**0.41**	**0.34**	**0.26**	*0.90*

[1] Baseline results are taken from [44].

TABLE XI. ATE (M) COMPARISON ON REPRESENTATIVE MARS-LVIG [43] SEQUENCES[1]. SEQUENCE HEADERS LIST FLIGHT SPEED AND ALTITUDE.

Method / Seq. Info.	Avg. Rank / RMSE	HKairport01 3 m/s / 80 m	HKairport02 6 m/s / 80 m	HKisland03 9 m/s / 90 m	AMtown03 12 m/s / 80 m	AMvalley03 12 m/s / 130 m	HKGNSS02 6 m/s / 80 m
LIO-Livox [70]	6.0/96.17	0.65	123.39	✗	✗	✗	2.97
FAST-LIO2 [23]	4.3/2.98	*0.44*	0.96	2.13	3.65	7.77	2.92
iG-LIO [95]	6.8/125.59	✗	✗	✗	3.54	✗	✗
R3LIVE [3]	5.3/51.40	0.68	*0.82*	3.93	✗	✗	2.98
AKF-LIO [96]	3.0/2.15	**0.43**	0.87	2.03	**2.74**	3.91	2.94
FAST-LIVO [25]	7.3/150.00	✗	✗	✗	✗	✗	✗
FAST-LIVO2 [15]	*2.7/1.42*	0.75	0.88	*0.89*	3.25	**1.43**	*1.32*
Ground-Fusion [7]	7.3/150.00	✗	✗	✗	✗	✗	✗
Ground-Fusion++ [18]	4.5/2.22	1.10	1.28	1.71	3.82	3.33	2.05
Ultra-Fusion (LVIO)	**1.3/1.40**	**0.43**	**0.61**	**0.87**	*3.01*	*2.60*	**0.90**

[1] Baseline results are taken from [43], [96], [97].

G. Mapping Results

Map consistency is evaluated qualitatively using the hybrid local map. Fig. 10 shows the MARS-LVIG HKisland03 reconstruction, where island contours, road boundaries, buildings, and terrain transitions remain coherent over a large-scale trajectory. The enlarged views show clear structural edges and limited ghosting in both intensity- and RGB-colored point clouds, supporting the use of optimized poses and aligned colorized LiDAR observations for Gaussian mapping.

H. Runtime Analysis

Runtime is profiled on 60 s segments with Robosense, Velodyne, Livox, and Hesai LiDARs using a real-time configuration: per-frame LiDAR frontend, window size 4, at most 3 nonlinear iterations, 12 ms solver budget, and capped factors on an Intel Core i9-14900K CPU. Ultra-Fusion requires 5.48–10.73 ms per optimization step (Fig. 11), satisfying real-time operation in this setting.

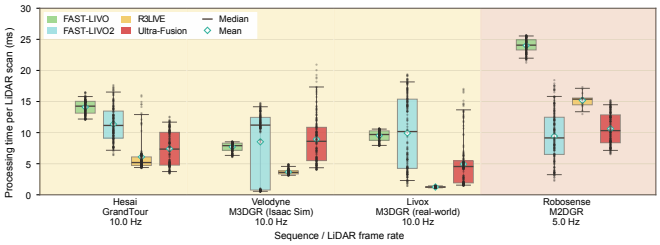

Fig. 11. Time consumption per LiDAR scan of Ultra-Fusion with baselines on different LiDAR types.

VI. CONCLUSION AND LIMITATIONS

This paper addresses multi-sensor localization for intelligent transportation systems under sensor degradation and spatiotemporal uncertainty. Ultra-Fusion formulates heterogeneous LiDAR, visual, inertial, wheel, and GNSS measurements as factors in a Unified Sliding-Window Estimator, so WIO, VIO, LIO, LVIO, and augmented configurations share state, calibration, reliability scheduling, and marginalization logic. Observability-Aware Initialization, Factor-Wise Reliability Scheduling, and Online Spatiotemporal Calibration improve initialization reliability, degradation tolerance, and calibration robustness. Extensive evaluation on M3DGR and public benchmarks shows competitive performance across wheeled, quadruped, and UAV platforms under long-term/high-speed motion and controlled perturbations.

Limitations: the study focuses on localization and geometric mapping; semantic scene understanding and explicit dynamic-object reasoning remain outside the evaluated scope. Extending the reliability-aware formulation toward semantic and dynamic environments is an important direction for future transportation deployments.

DECLARATIONS AND DATA AVAILABILITY

Data Availability. Most M3DGR sequences are publicly available at https://github.com/sjtuyinjie/M3DGR. The two long-duration sequences and M3DGR Sim sequences will be released upon acceptance. Executable binaries of Ultra-Fusion are available at https://github.com/sjtuyinjie/Ultra-Fusion; source code will be released upon acceptance.

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SUPPLEMENTARY MATERIALS

In this material, we provide supplementary tables for system configuration, benchmark comparison, and evaluated-framework licensing information.

A. System Illustration

Fig. 12 gives a runtime-level illustration of Ultra-Fusion. The system first aligns asynchronous ROS topics in a timestamp-ordered buffer, then converts the admitted measurements into modality-specific front-end products. Observability-Aware Initialization (OAI), Factor-Wise Reliability Scheduling (FRS), and Online Spatiotemporal Calibration (OSC) operate before or alongside the common backend, so WIO, VIO, LIO, LVIO, and optional wheel/GNSS-augmented modes share the same state, marginalization prior, and solver interface.

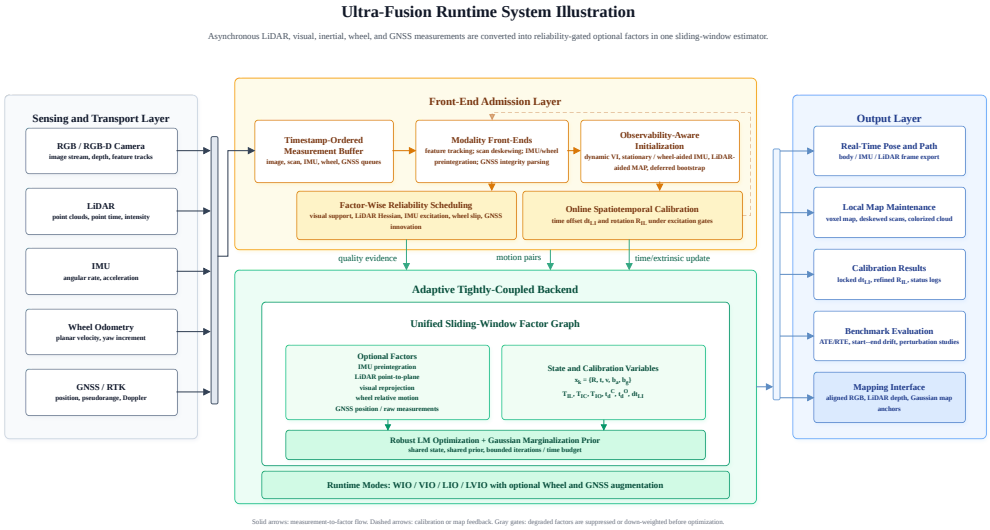

Fig. 12. System illustration of Ultra-Fusion. Heterogeneous measurements are timestamp-ordered, screened by modality front-ends, initialized under observability-aware bootstrap, and admitted into a unified sliding-window factor graph through reliability-gated optional factors. The same backend state supports pose estimation, local map maintenance, online LiDAR–IMU spatiotemporal calibration, and benchmark outputs across WIO, VIO, LIO, and LVIO configurations.

B. Key Parameters

Table XII maps the runtime blocks in Fig. 12 to the main configuration parameters used by Ultra-Fusion. The listed values follow the default M3DGR configuration; sequence-specific files may adjust noise levels, residual caps, and platform-dependent gates for different LiDAR types or motion regimes.

C. M3DGR Dataset

Sequence details. The real-world M3DGR sequences cover routine operation and four transportation-relevant corner cases. For visual degradation, we consider dim lighting, time-varying illumination, dynamic foreground motion, and partial or complete occlusion. Indoors, darkness is emulated by switching off room lights and using a phone flashlight; illumination variation is created by periodically toggling the lighting condition; dynamic interference is induced by having a person move through the camera field of view; and occlusion is generated by intentionally blocking the lens. Outdoors, low-light and illumination-change sequences are recorded at night, while dynamic scenes include pedestrians, cyclists, and vehicles.

For LiDAR degeneracy, two sequence types emphasize poor geometric observability: a long corridor and a corridor-to-elevator transition. In both settings, the robot executes a loop and returns to the starting area, and ArUco-based start–end alignment is used to quantify accumulated drift caused by weak structural constraints. For wheel slippage, four corner cases are included: suspended-wheel “float” events, sharp turns, low-traction grass traversal, and rough roads with abrupt elevation changes. These sequences stress whether a fusion system can suppress corrupted wheel measurements before they bias the estimator. The GNSS-denial sequence starts with stable satellite reception, traverses a region without GNSS coverage, and returns to the initial area; ArUco markers make the induced drift observable even when absolute GNSS updates are absent.

Ground truth uses motion capture indoors at 360 Hz and RTK GNSS outdoors at 15 Hz, with ArUco-based relative alignment used for start–end drift evaluation in degeneracy and GNSS-denial cases. The M3DGR Sim sequences provide simulator ground truth, including exact sensor extrinsics, and are grouped into Wild, Warehouse, and Tunnel scenes for controlled perturbation analysis. Table XIV summarizes the number of sequences, distance, duration, storage size, and ground-truth source for each scenario category.

TABLE XII. KEY RUNTIME PARAMETERS OF ULTRA-FUSION.

Group / block	Parameters	Runtime role	Default values
Input and front-end processing			
Sensor buffer	Topic association, packet ordering, point filtering	Provides timestamp-ordered packets for all enabled modalities	IMU 100 Hz; image 10 Hz; wheel 10 Hz; LiDAR blind range 0.1 m
Visual front-end	Feature count, spacing, track length, parallax	Produces visual tracks and reprojection factors when visual support is sufficient	Max 150 features; min distance 30 px; min track length 4; keyframe parallax 10 px
LiDAR front-end / map	Residual cap, voxel size, plane gate, robust loss	Produces scan-to-map factors and LiDAR reliability evidence	Residual cap 1200–2000; voxel 0.5 m; Huber $\sigma = 0.5$–1.0; plane eigen-ratio gate 0.12
Motion support and global anchoring			
IMU / wheel support	IMU noise, wheel noise, stationary checks	Supplies preintegration and slip/stationary evidence	$\sigma_a = 0.04$, $\sigma_g = 0.008$; wheel velocity noise 0.2–0.6; gravity alignment 1.0 s
GNSS anchoring	Satellite/elevation checks, covariance, innovation gate	Admits optional global factor after integrity checks	Elevation 30°; min satellites 5; innovation gate 5.0; horizontal/vertical noise 5/20 m
Robustness and calibration			
OAI bootstrap	Initial LiDAR frames, IMU window, wheel-aided checks	Selects visual–inertial, stationary/wheel-aided, LiDAR-aided MAP, or deferred initialization	LiDAR init frames 20; IMU gravity init 1.0 s; wheel stop threshold 0.01
FRS gates	LiDAR geometry, visual support, IMU excitation, wheel slip, GNSS innovation	Activates, suppresses, or covariance-inflates factors before backend optimization	Visual min feature 100; LiDAR normal radius 0.5 m; sigma gate $k = 3.0$
OSC worker	Time-offset search, overlap duration, confidence, extrinsic lock	Updates LiDAR–IMU time association and rotation under sufficient excitation	Δt_{LI} range ± 0.5 s; 12 s window; min overlap 5 s; confidence 0.6
Optimization backend			
Unified backend	Window size, state variables, optional factors, solver budget	Jointly optimizes active WIO/VIO/LIO/LVIO factors with shared marginalization	10-state window; Ceres LM; 8 iterations; 40 ms budget

TABLE XIII. COMPARISON OF REPRESENTATIVE SLAM BENCHMARK DATASETS IN TERMS OF SCENARIOS, SENSORS, AND EVALUATION BREADTH.

Dataset/Year	Scenario				Sensors							Number of compared algorithms
	VC[1]	LR[2]	GD[3]	WS[4]	RGB	Depth	Omni[5]	IMU	LiDAR	Wheel	GNSS	
EuRoC, 2016	✓				✓			✓				0
KAIST, 2019	✓	✓			✓			✓	4	✓	✓	0
UrbanLoco, 2020	✓		✓		✓			✓	1		✓	3
OpenLoris-Scene, 2020	✓			✓	✓	✓		✓		✓		9
M2DGR, 2021	✓	✓	✓		✓		✓	✓	1		✓	10
FusionPortable, 2022	✓				✓			✓	1		✓	5
Ground-Challenge, 2023	✓	✓		✓	✓	✓		✓				5
M2DGR-Plus, 2024	✓		✓	✓	✓	✓		✓	1	✓	✓	6
MARS-LVIG, 2024	✓	✓			✓			✓	1		✓	6
GrandTour, 2026	✓	✓	✓		✓	✓		✓	3		✓	*52*
M3DGR (Ours), 2026	✓	✓	✓	✓	✓	✓	✓	✓	2	✓	✓	**68**

[1] is visual challenge, [2] is LiDAR degeneracy, [3] is GNSS denied zone, [4] is wheel slippage and [5] is omnidirectional camera.

TABLE XIV. OVERVIEW OF SCENARIOS IN THE M3DGR BENCHMARK, INCLUDING REAL-WORLD AND SIMULATION SEQUENCES.

Scenario	Visual Challenge				LiDAR Degeneracy		Wheel Slippage				GNSS Denial	Standard	Sim Data			TOTAL
	Dark	VI[1]	Dynamic	Occlusion	Corridor	Elevator	WF[2]	ST[3]	Grass	RR[4]			Wild	Warehouse	Tunnel	
Number	5	4	3	4	2	1	2	2	2	1	2	4	2	1	2	37
Dist/m	1653.31	1055.58	355.97	1091.24	545.64	470.64	101.55	170.88	318.91	457.35	1162.39	4485.49	159.991	10.123	861.257	12900.321
Duration/s	2274	1458	609	1224	696	699	171	238	459	533	1359	5101	254.980	194.990	1053.480	16324.45
Size/GB	27.0	20.0	7.1	12.3	11.9	11.2	3.3	2.9	9.7	10.4	23.2	86.0	14.4	2.6	60.4	302.4
Ground Truth	RTK/Mocap	RTK/Mocap	RTK/Mocap	RTK/Mocap	ArUco	ArUco	Mocap	Mocap	RTK	RTK	ArUco	RTK	Sim	Sim	Sim	—

[1] VI: varying illumination. [2] WF: wheel float. [3] ST: sharp turn. [4] RR: rough road.

D. Evaluated Frameworks and Licenses

Table XV lists the frameworks evaluated in the M3DGR benchmark comparison of the main paper, together with their tested categories, public repository links, and associated open-source licenses when available. Methods follow the same order as Table I in the main paper.

TABLE XV. EVALUATED FRAMEWORKS AND THEIR ASSOCIATED LICENSES.

Method	Tested Category	Repository Link	License
Ground-Fusion WIO	WIO	https://github.com/SJTU-ViSYS/Ground-Fusion	GPL-3.0
GNSS SPP	GNSS reference	N/A	N/A
Ultra-Fusion (WIO)	WIO	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
ORB-SLAM2	VO	https://github.com/raulmur/ORB_SLAM2	NOASSERTION
VINS-Mono	VIO	https://github.com/HKUST-Aerial-Robotics/VINS-Mono	GPL-3.0
VINS-RGBD	RGBD-VIO	https://github.com/Lab-of-AI-and-Robotics/VINS-RGBD	GPL-3.0
TartanVO	VO	https://github.com/castacks/tartanvo	BSD-3-Clause
ORB-SLAM3	VO	https://github.com/UZ-SLAMLab/ORB-SLAM3	GPL-3.0
VINS-GPS-Wheel	VIO+Wheel+GNSS	https://github.com/Wallong/VINS-GPS-Wheel	GPL-3.0
DM-VIO	VIO	https://github.com/lukasvst/dm-vio	GPL-3.0
GVINS	GVIO	https://github.com/HKUST-Aerial-Robotics/GVINS	GPL-3.0
VIW-Fusion	VIW	https://github.com/TouchDeeper/VIW-Fusion	GPL-3.0
Ground-Fusion	Multi-sensor	https://github.com/SJTU-ViSYS/Ground-Fusion	GPL-3.0
MASt3R-SLAM	VO	https://github.com/rmurai0610/MASt3R-SLAM	NOASSERTION
Ultra-Fusion (VIO)	VIO	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
Ultra-Fusion (VWIO)	VIW	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
A-LOAM	LO	https://github.com/HKUST-Aerial-Robotics/A-LOAM	NOASSERTION
LeGO-LOAM	LO	https://github.com/RobustFieldAutonomyLab/LeGO-LOAM	BSD-3-Clause
LIO-mapping	LIO	https://github.com/hyye/lio-mapping	GPL-3.0
LIO-SAM	LIO	https://github.com/TixiaoShan/LIO-SAM	BSD-3-Clause
LINS	LIO	https://github.com/ChaoqinRobotics/LINS---LiDAR-inertial-SLAM	NOASSERTION
LOAM-Livox	LO	https://github.com/hku-mars/loam_livox	GPL-2.0
LiLi-OM	LIO	https://github.com/KIT-ISAS/lili-om	GPL-3.0
LIO-Livox	LIO	https://github.com/Livox-SDK/livox_mapping	NOASSERTION
Faster-LIO	LIO	https://github.com/gaoxiang12/faster-lio	GPL-2.0
IESKF-LIO	LIO	https://github.com/chengwei0427/ESKF_LIO	NOASSERTION
VoxelMap	LIO	https://github.com/hku-mars/VoxelMap	NOASSERTION
Fast-LIO2	LIO	https://github.com/hku-mars/FAST_LIO	GPL-2.0
CTLO	LO	https://github.com/G3tupup/ctlo	Unknown
Point-LIO	LIO	https://github.com/hku-mars/Point-LIO	Custom
LOG-LIO	LIO	https://github.com/tiev-tongji/LOG-LIO	GPL-2.0
CT-LIO	LIO	https://github.com/chengwei0427/ct-lio	GPL-2.0
DLIO	LIO	https://github.com/vectr-ucla/direct_lidar_inertial_odometry	MIT
HM-LIO	LIO	https://github.com/chengwei0427/hm-lio	NOASSERTION
KISS-ICP	LO	https://github.com/PRBonn/kiss-icp	MIT
SLICT	LIO	https://github.com/brytsknguyen/slict	GPL-2.0
MM-LINS	LIO	https://github.com/lian-yue0515/MM-LINS	NOASSERTION
SLICT2	LIO	https://github.com/brytsknguyen/slict	GPL-2.0
PIN-SLAM	LiDAR-SLAM	https://github.com/PRBonn/PIN_SLAM	MIT
I2EKF-LO	LO	https://github.com/YWL0720/I2EKF-LO	GPL-2.0
LTAOM	LO	https://github.com/hku-mars/LTAOM	NOASSERTION
LOG-LIO2	LIO	https://github.com/tiev-tongji/LOG-LIO2	GPL-2.0
Eq-LIO	LIO	https://github.com/Eliaul/Eq-LIO	NOASSERTION
Traj-LO	LO	https://github.com/kevin2431/Traj-LO	MIT
VoxelMap++	LIO	https://github.com/uestc-icsp/VoxelMapPlus_Public	NOASSERTION
DMSA-SLAM	LiDAR-SLAM	https://github.com/davidskdds/DMSA_LiDAR_SLAM	MIT
Adaptive-LIO	LIO	https://github.com/chengwei0427/Adaptive-LIO	BSD-3-Clause
GLO	LO	https://github.com/robosu12/GLO	GPL-3.0
LIGO	LIO+GNSS	https://github.com/Joanna-HE/LIGO.	BSD-3-Clause
CTE-MLO	(M)LO	https://github.com/shenhm516/CTE-MLO	GPL-2.0
RKO-LIO	LIO	https://github.com/PRBonn/rko_lio	MIT
II-NVM	LIO	https://github.com/chengwei0427/II-NVM	NOASSERTION
Surfel-LIO	LIO	https://github.com/93won/lidar_inertial_odometry	MIT
GenZ-ICP	LO	https://github.com/cocel-postech/genz-icp	MIT
Voxel-SLAM	LiDAR-SLAM	https://github.com/hku-mars/Voxel-SLAM	GPL-2.0
Ultra-Fusion (LIO)	LIO	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
Ultra-Fusion (LWIO)	LIO+Wheel	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
LVI-SAM	LIVO	https://github.com/TixiaoShan/LVI-SAM	BSD-3-Clause
R2LIVE	LIVO	https://github.com/hku-mars/r2live	GPL-2.0
R3LIVE	LIVO	https://github.com/hku-mars/r3live	GPL-2.0
Fast-LIVO	LIVO	https://github.com/hku-mars/FAST-LIVO	GPL-2.0
Coco-LIC	LIVO	https://github.com/APRIL-ZJU/Coco-LIC	GPL-3.0
SR-LIVO	LIVO	https://github.com/ZikangYuan/sr_livo	GPL-2.0
Fast-LIVO2	LIVO	https://github.com/hku-mars/FAST-LIVO2	GPL-2.0
Ground-Fusion++	Multi-sensor	https://github.com/sjtuyinjie/Ground-Fusion2	MIT
Ultra-Fusion (LVIO)	LVIO	https://github.com/sjtuyinjie/Ultra-Fusion	MIT
Ultra-Fusion (LVWIO)	LVWIO	https://github.com/sjtuyinjie/Ultra-Fusion	MIT